

Generalized linear model



02/06/2026

Logistics

Midterm

- will be released shortly after today's class
- midterm is take-home, open exam (you can use all the course notes, all of the internet, etc.)
- is due on **Friday, February 13th at 8pm** (stretch days can't be used for the midterm)
- midterm is like the other homeworks but:
 - it's longer
 - you'll need to work on it on your own
- there won't be office hours or sections in the week that the midterm is due
- **no class on Wednesday** next week
- ask questions about the midterm on Ed Discussion (please **ask private first** -- we will then respond and, if appropriate, check with you whether we can share with the rest of the class)

Midterm

Introduction

This is a take-home exam. The exam is open notes and open book (in short, you can use any source of information you like as long as you work on the exam by yourself). The maximum score is 120 points. Please adhere to the honor code. Submit the midterm as a PDF on the canvas ‘midterm’ assignment by **Friday, February 14th, 8pm**.

Upload both the rendered pdf file (252_midterm.pdf) and the Rmarkdown code file (252_midterm.rmd) to Canvas.

The late policy submission policy is:

- We will subtract 2% from your points for each hour that the midterm is submitted late but before midnight. For example, 2% will be subtracted if you submit between 8pm and 9pm, or 8% if you submit between 11pm and midnight.
- 20% will be subtracted if you submit after midnight on Friday but before 8pm on Saturday, February 15th.
- No points will be granted if you submit later than 8pm on Saturday, February 15th.

For questions that require written responses, please make sure to show any relevant tables, summaries (e.g. from `lm()` or `anova()`), or visualizations. Some of the code chunks have existing code that you can use to build your code around.

When asked to report results, please do so like you would in a scientific article (see examples from lectures, as well as in **‘Reporting Results.pdf’** on Canvas under **Files > papers**).

- Please leave the `\clearpage` commands where they are. This makes sure that each question is printed on a separate page in the pdf.
- Some code chunks are set to `eval=F`, make sure to set these to `eval=T` before knitting the final version.
- We note for each question how many points you can get. You can get up to 120 points in total.
- Good coding style matters! We will add or subtract up to 5 points depending on style.
- **Pro tip:** Read each question carefully so you don’t miss any instructions. There are sometimes multiple sub-tasks or questions, and try to answer them all.

If you have any questions about the midterm, please post them on Ed Discussion addressed to the instructors only. We will answer your question and check with you whether we can share both your question and our answer with the rest of the group.

Best of luck with the midterm!

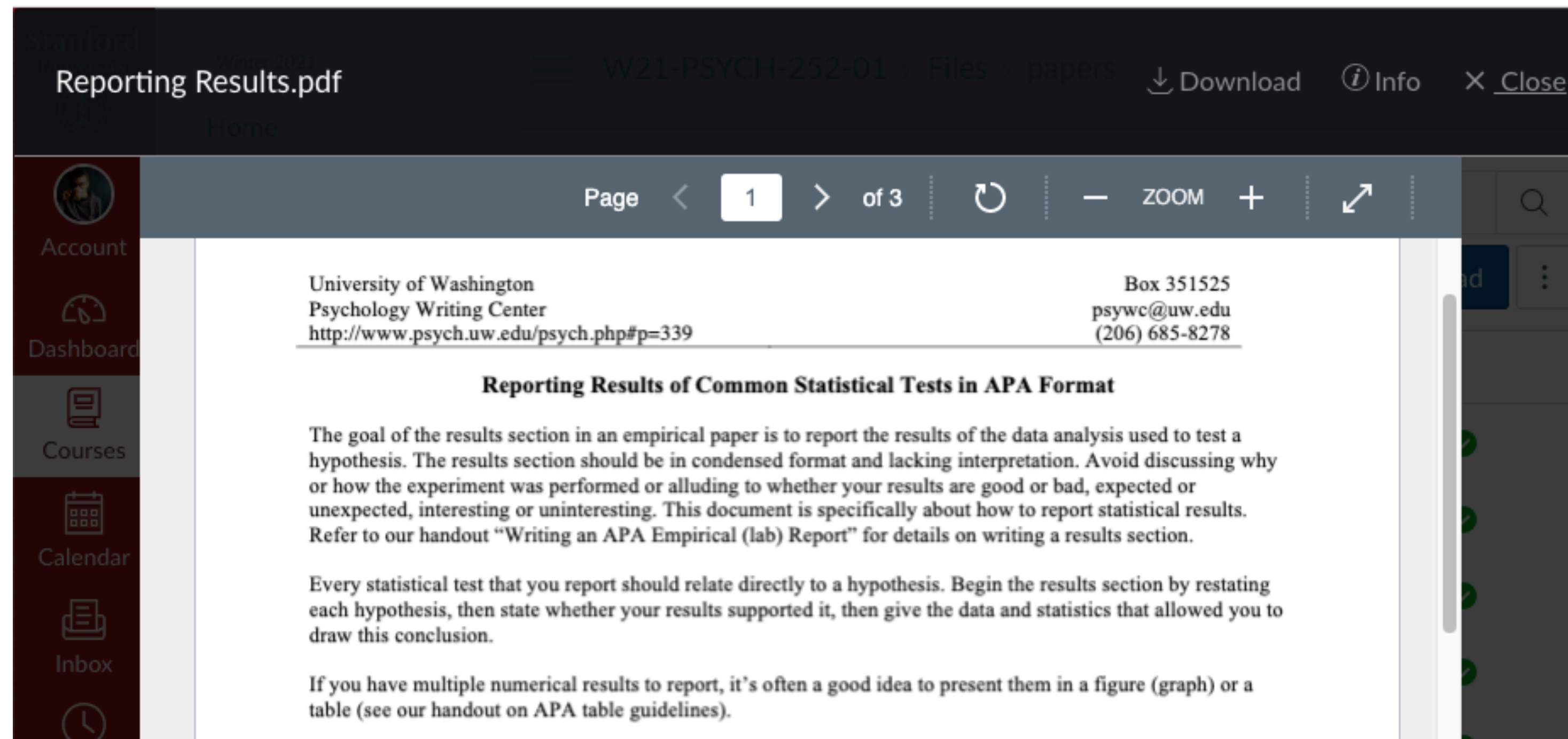
The Honor Code is the University’s statement on academic integrity written by students in 1921. It articulates University expectations of students and faculty in establishing and maintaining the highest standards in academic work:

1. The Honor Code is an undertaking of the students, individually and collectively:
 - a. that they will not give or receive aid in examinations; that they will not give or receive unpermitted aid in class work, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;
 - b. that they will do their share and take an active part in seeing to it that others as well as themselves uphold the spirit and letter of the Honor Code.
2. The faculty on its part manifests its confidence in the honor of its students by refraining from proctoring examinations and from taking unusual and unreasonable precautions to prevent the forms of dishonesty mentioned above. The faculty will also avoid, as far as practicable, academic procedures that create temptations to violate the Honor Code.
3. While the faculty alone has the right and obligation to set academic requirements, the students and faculty will work together to establish optimal conditions for honorable academic work.

3 parts, 40 points each

Reporting results

- When we ask you to report results in the homework, try and do this like you would do in a scientific paper (reporting the statistical test, test-statistic, p-value, effect size).



Files > papers > Reporting results.pdf

Plan for today

- Quick Recap
- Interpreting parameters
- Unbalanced designs
- Linear contrasts
- Generalized linear model
 - Logistic regression
 - interpreting the model output
 - fitting and reporting models

Quick recap

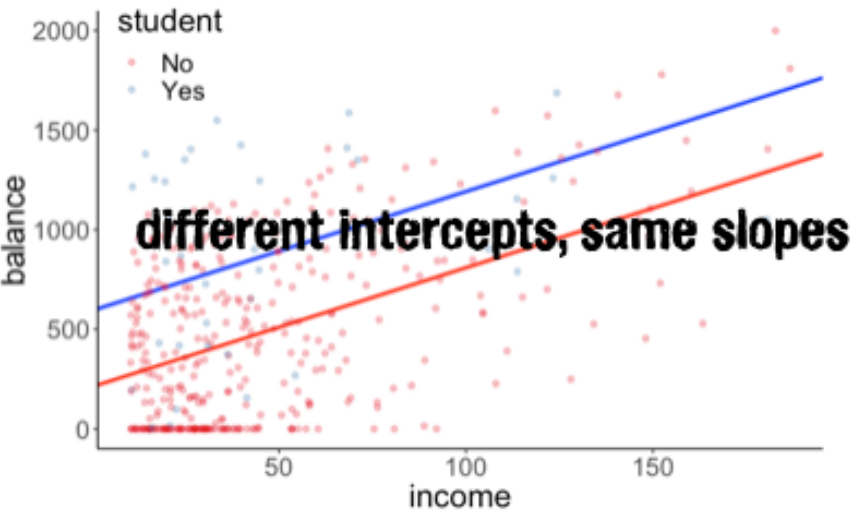
Quick recap: Interactions

H_0 : The relationship between income and balance is the same for students and non-students.

Model C

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{student}_i + \epsilon_i$$

Model prediction



Fitted model

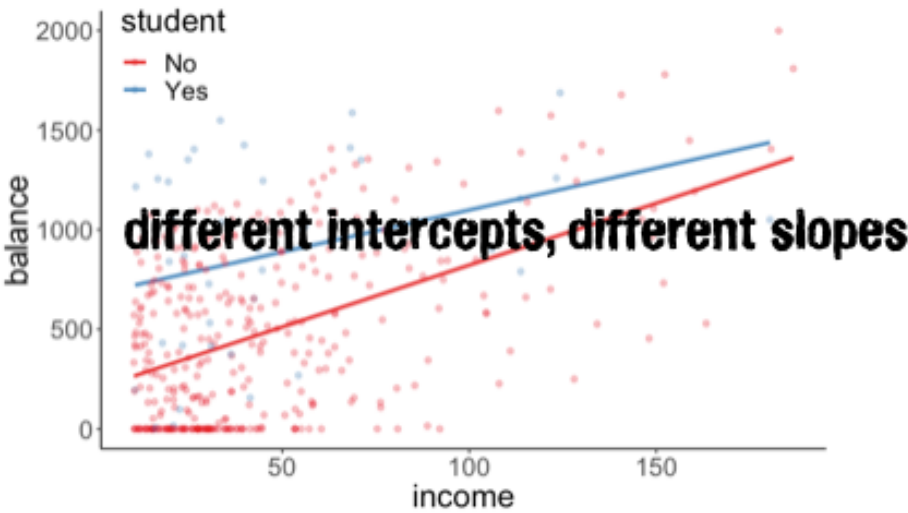
$$\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + 382.67 \cdot \text{student}_i$$

H_1 : The relationship between income and balance differs between students and non-students.

Model A

$$\widehat{\text{balance}}_i = b_0 + b_1 \text{income}_i + b_2 \text{student}_i + b_3(\text{income}_i \times \text{student}_i) + \epsilon_i$$

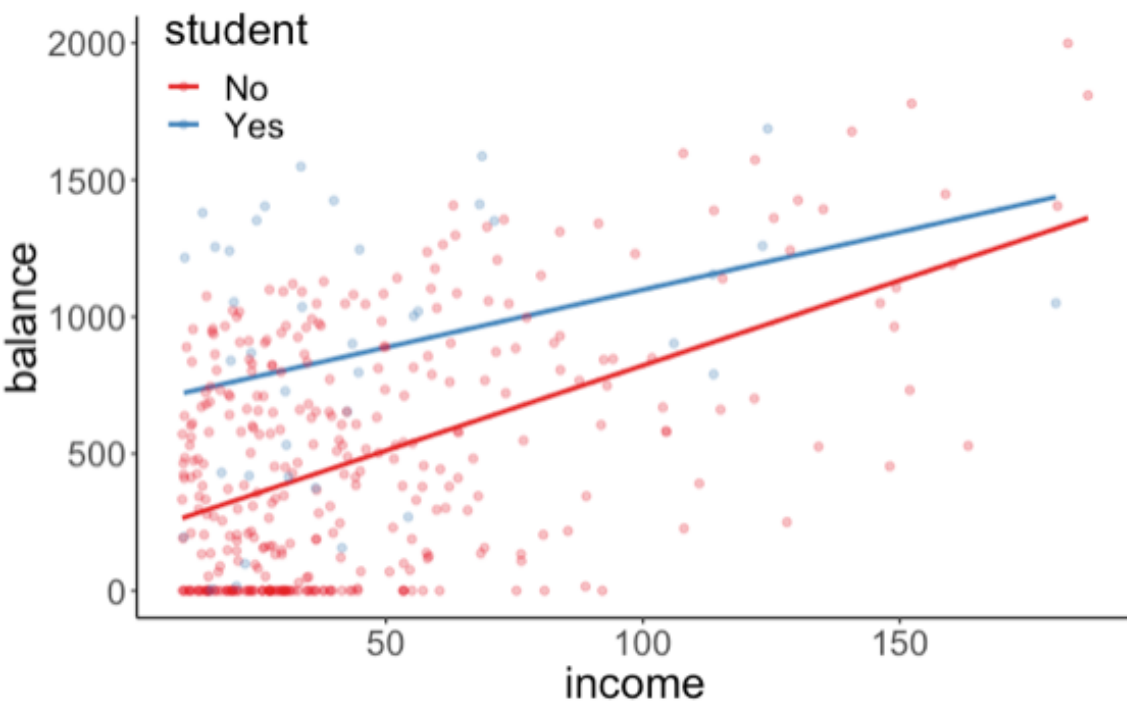
Model prediction



Fitted model

$$\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot \text{student}_i - 2.00 \cdot (\text{income}_i \times \text{student}_i)$$

Interpretation



$$\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot \text{student}_i - 2.00 \cdot (\text{income}_i \times \text{student}_i) + 0$$

if student = "No" $\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + (476.68 \cdot 0) - 2.00 \cdot (\text{income}_i \times 0)$

if student = "Yes"

$$\begin{aligned} \widehat{\text{balance}}_i &= 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot 1 - 2.00 \cdot (\text{income}_i \times 1) \\ &= 677.3 + 6.22 \cdot \text{income}_i - 2.00 \cdot \text{income}_i \\ &= 677.3 + 4.22 \cdot \text{income}_i \end{aligned}$$

22

```
Call:
lm(formula = balance ~ 1 + income + student, data = df.credit)

Residuals:
    Min       1Q   Median       3Q      Max
-762.37 -331.38  -45.04   323.60   818.28

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  211.1430    32.4572   6.505 2.34e-10 ***
income         5.9843     0.5566  10.751 < 2e-16 ***
studentYes   382.6705    65.3108   5.859 9.78e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.8 on 397 degrees of freedom
Multiple R-squared:  0.2775, Adjusted R-squared:  0.2738
F-statistic: 76.22 on 2 and 397 DF, p-value: < 2.2e-16
```

effect of income
for non-students

```
Call:
lm(formula = balance ~ income + student + income:student, data = df.credit)

Residuals:
    Min       1Q   Median       3Q      Max
-773.39 -325.70  -41.13   321.65   814.04

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  200.6232    33.6984   5.953 5.79e-09 ***
income         6.2182     0.5921  10.502 < 2e-16 ***
studentYes   476.6758   104.3512   4.568 6.59e-06 ***
income:studentYes -1.9992     1.7313  -1.155  0.249
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.6 on 396 degrees of freedom
Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744
F-statistic: 51.3 on 3 and 396 DF, p-value: < 2.2e-16
```

effect of income
for non-students

Quick recap: `lm()` output

```
1 lm(formula = balance ~ income + student + income:student, data = df.credit) %>%  
2 summary()
```

```
Call:  
lm(formula = balance ~ income + student + income:student,  
    data = df.credit)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-773.39 -325.70  -41.13   321.65   814.04   
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)    200.6232    33.6984   5.953 5.79e-09 ***  
income           6.2182     0.5921  10.502 < 2e-16 ***  
studentYes     476.6758   104.3512   4.568 6.59e-06 ***  
income:studentYes -1.9992     1.7313  -1.155  0.249      
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 391.6 on 396 degrees of freedom  
Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744   
F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

assesses a specific predictor

```
1 fit_c = lm(formula = balance ~ student + income:student, data = df.credit)  
2 fit_a = lm(formula = balance ~ income + student + income:student, data = df.credit)  
3  
4 anova(fit_c, fit_a)
```

assesses the
full model

```
1 fit_c = lm(formula = balance ~ 1, data = df.credit)  
2 fit_a = lm(formula = balance ~ 1 + income + student + income:student, data = df.credit)  
3  
4 anova(fit_c, fit_a)
```

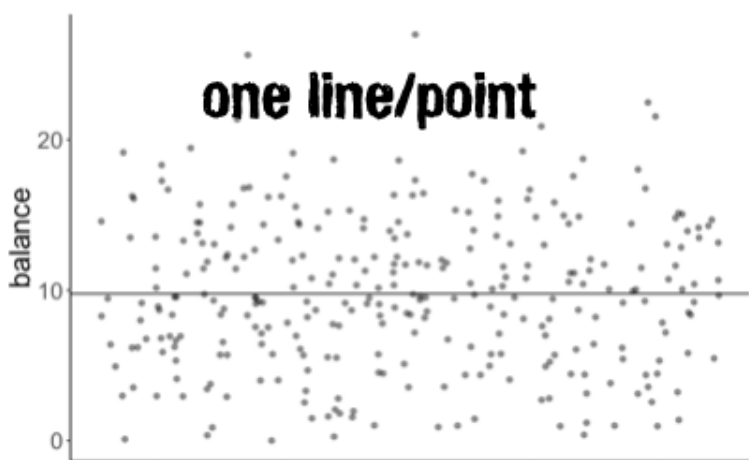
Quick recap: Categorical variables (single predictor)

H₀: Card quality does not affect the final balance.

Model C

$$\text{balance}_i = \beta_0 + \epsilon_i$$

Model prediction



Fitted model

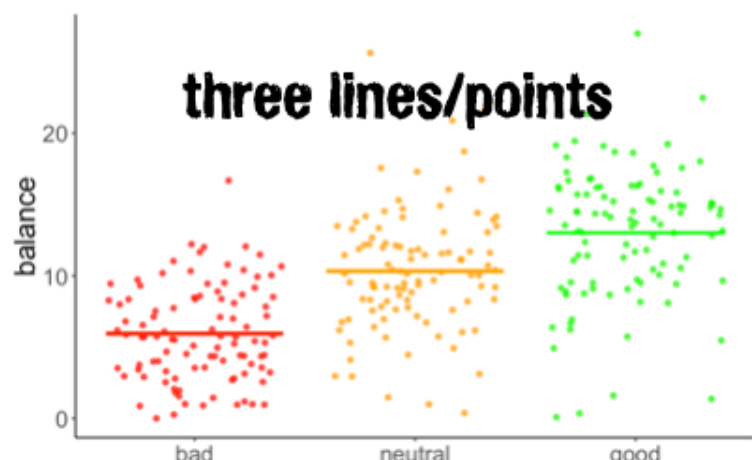
$$\widehat{\text{balance}}_i = 9.77$$

H₁: Card quality affects the final balance.

Model A

$$\text{balance}_i = \beta_0 + \beta_1 \text{hand_neutral}_i + \beta_2 \text{hand_good}_i + \epsilon$$

Model prediction



Fitted model

$$\widehat{\text{balance}}_i = 5.94 + 4.41 \cdot \text{hand_neutral}_i + 7.08 \cdot \text{hand_good}_i$$

```
lm(formula = balance ~ 1 + hand, data = df.poker)
```

```
Call:
lm(formula = balance ~ hand, data = df.poker)

Residuals:
    Min       1Q   Median       3Q      Max
-12.9264  -2.5902  -0.0115   2.6573  15.2834

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.9415    0.4111  14.451  < 2e-16 ***
handneutral   4.4051    0.5815   7.576 4.55e-13 ***
handgood      7.0849    0.5815  12.185  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

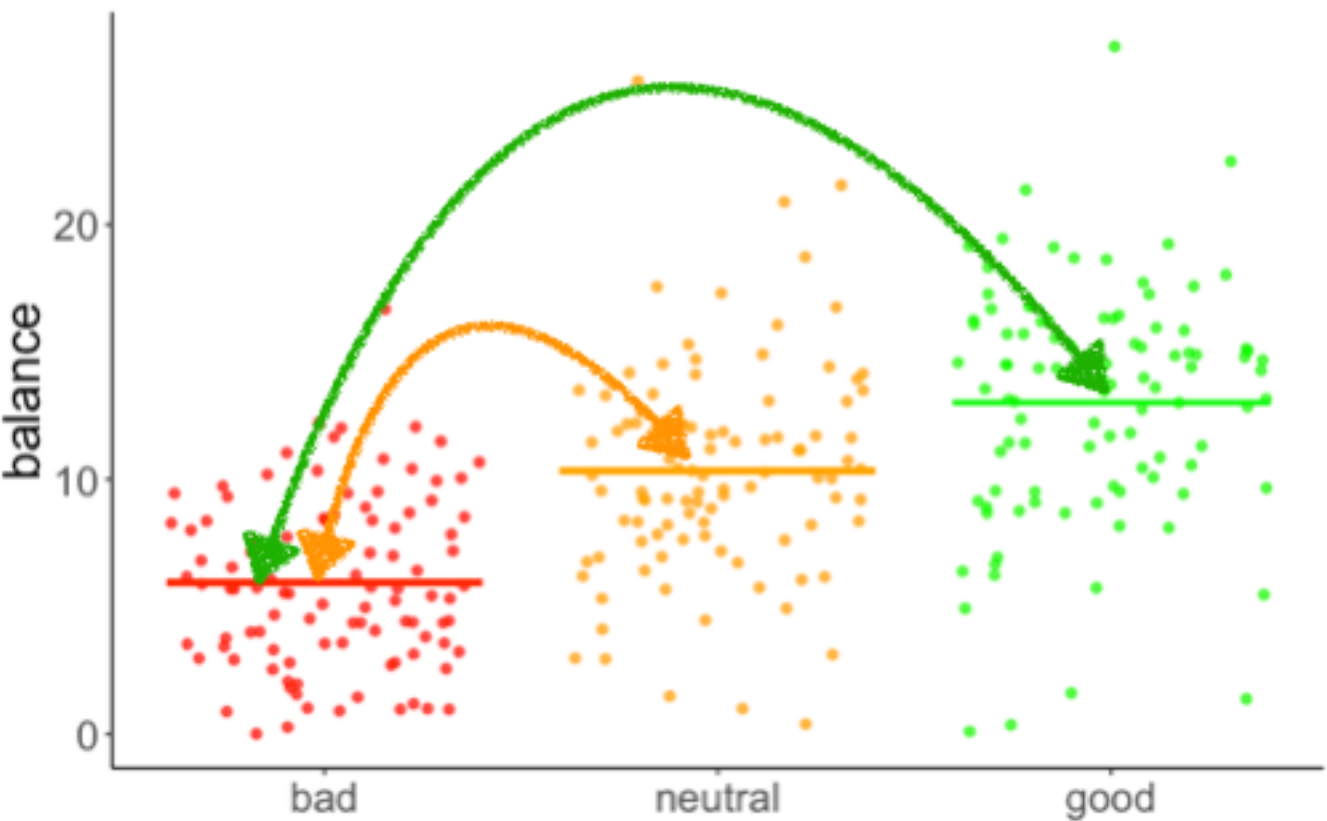
Residual standard error: 4.111 on 297 degrees of freedom
Multiple R-squared:  0.3377, Adjusted R-squared:  0.3332
F-statistic: 75.7 on 2 and 297 DF, p-value: < 2.2e-16
```

Interpreting the results

regression coefficients encode differences between group means

term	estimate	std.error	statistic	p.value
(Intercept)	5.941	0.411	14.451	0
handneutral	4.405	0.581	7.576	0
handgood	7.085	0.581	12.185	0

$$\widehat{\text{balance}}_i = 5.94 + 4.41 \cdot \text{hand_neutral}_i + 7.08 \cdot \text{hand_good}_i$$



if hand == "bad":

$$\widehat{\text{balance}}_i = 5.94$$

if hand == "neutral":

$$\widehat{\text{balance}}_i = 5.94 + 4.41 = 10.35$$

if hand == "good":

$$\widehat{\text{balance}}_i = 5.94 + 7.08 = 13.02$$

Factor Variables
Group Variables
Mean Differences (ANOVA)

Interpreting parameters

Parameter interpretation

```
lm(formula = balance ~ 1 + hand * skill, data = df.poker) %>%  
summary()
```

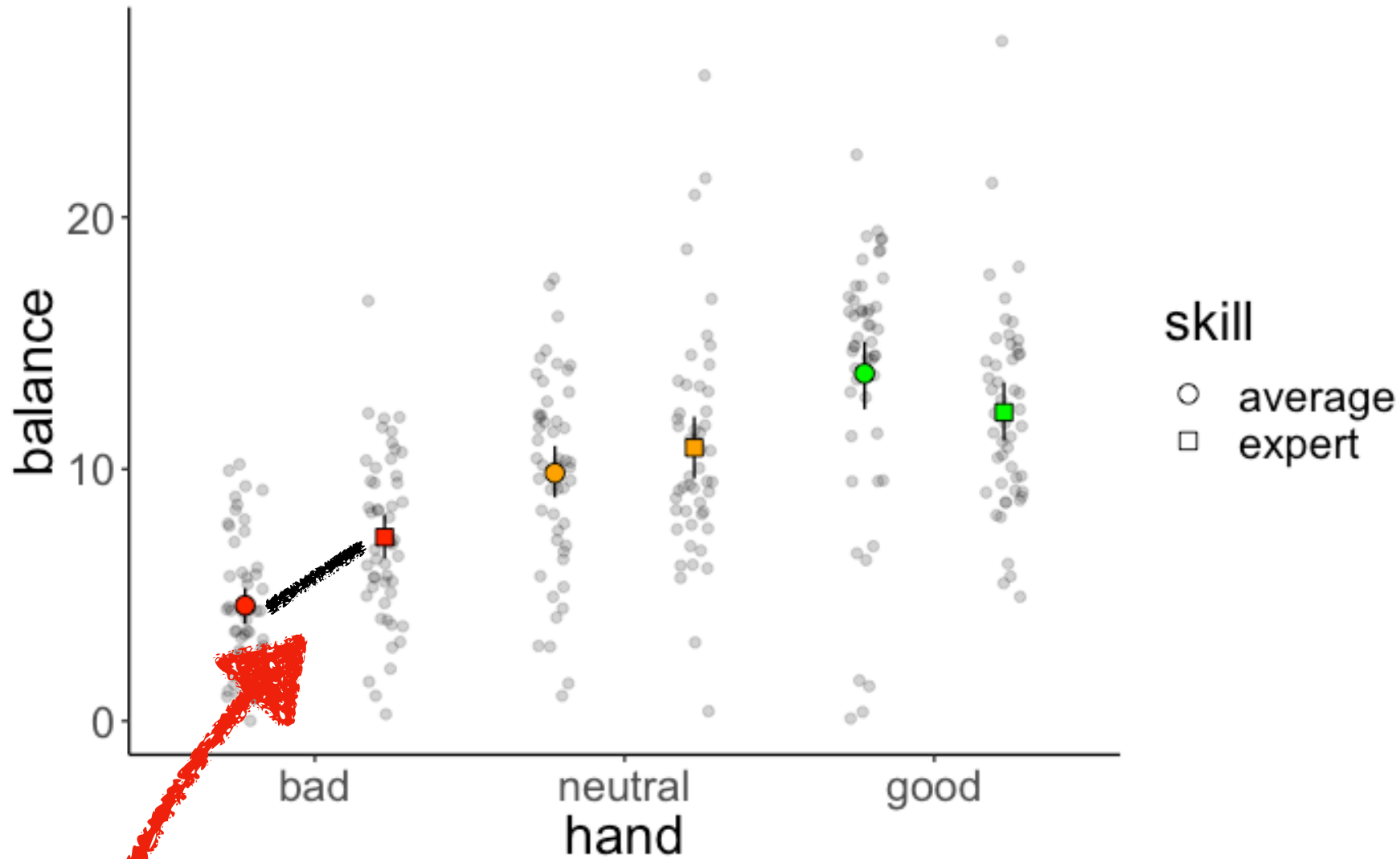
```
Call:  
lm(formula = balance ~ 1 + hand * skill, data = df.poker)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-13.6976  -2.4740   0.0348   2.4644  14.7806   
  
Coefficients:  
                Estimate Std. Error t value Pr(>|t|)      
(Intercept)      4.5866     0.5686   8.067 1.85e-14 ***  
handneutral       5.2572     0.8041   6.538 2.75e-10 ***  
handgood          9.2110     0.8041  11.455 < 2e-16 ***  
skillexpert       2.7098     0.8041   3.370 0.000852 ***  
handneutral:skillexpert -1.7042     1.1372  -1.499 0.135038   
handgood:skillexpert  -4.2522     1.1372  -3.739 0.000222 ***  
---  
Sign
```

acute danger of misinterpretation!!

Residual standard error: 4.02 on 294 degrees of freedom
Multiple R-squared: 0.3731, Adjusted R-squared: 0.3624
F-statistic: 34.99 on 5 and 294 DF, p-value: < 2.2e-16

~~there was a significant effect of skill~~

Parameter interpretation



```
Call:
lm(formula = balance ~ hand * skill, data = df.poker)

Residuals:
    Min       1Q   Median       3Q      Max
-13.6976  -2.4740   0.0348   2.4644  14.7806

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.5866     0.5686   8.067 1.85e-14 ***
handneutral      5.2572     0.8041   6.538 2.75e-10 ***
handgood         9.2110     0.8041  11.455 < 2e-16 ***
skillexpert      2.7098     0.8041   3.370 0.000852 ***
handneutral:skillexpert -1.7042     1.1372  -1.499 0.135038
handgood:skillexpert  -4.2522     1.1372  -3.739 0.000222 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

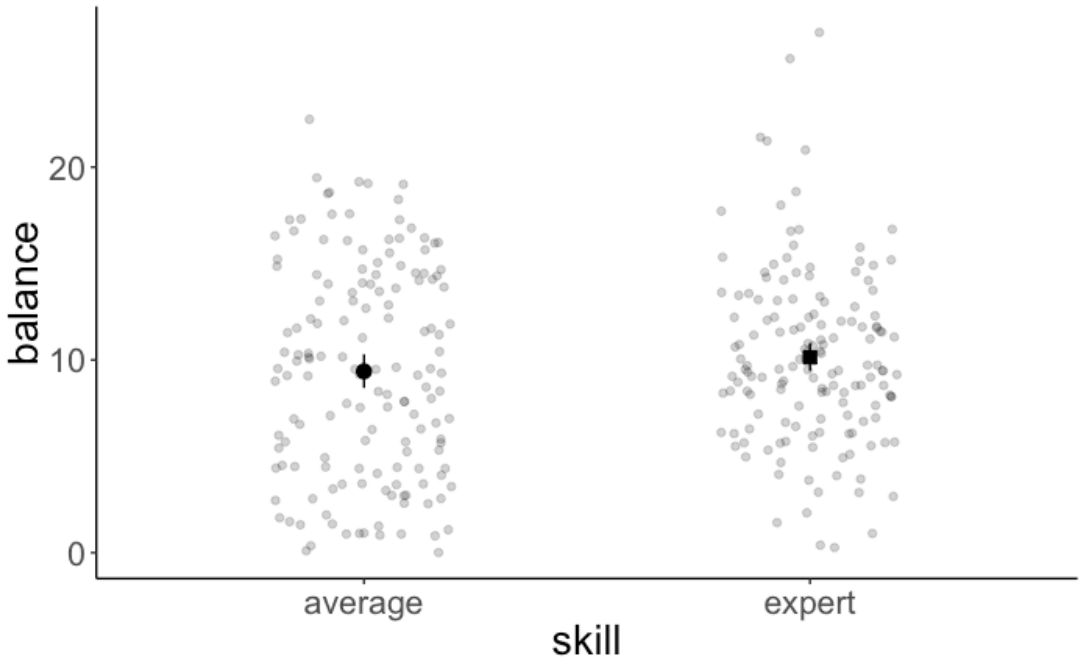
Residual standard error: 4.02 on 294 degrees of freedom
Multiple R-squared:  0.3731, Adjusted R-squared:  0.3624
F-statistic: 34.99 on 5 and 294 DF, p-value: < 2.2e-16
```

```
lm(formula = balance ~ hand * skill,
    data = df.poker) %>%
anova()
```

```
Analysis of Variance Table

Response: balance
      Df Sum Sq Mean Sq F value    Pr(>F)    
hand     2 2559.4  1279.70   79.1692 < 2.2e-16 ***
skill     1   39.3    39.35    2.4344 0.1197776
hand:skill 2   229.0   114.49    7.0830 0.0009901 ***
Residuals 294 4752.3    16.16                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

there was no main effect of skill!



is this difference significantly different from 0?

hand	average	expert	difference
bad	4.59	7.3	2.71

Different effect terms

- **main effect:** effect of one independent variable on the dependent variable
- **interaction effect:** when the effect of one independent variable depends on the level of another (**moderation**)
- **simple effect:** comparison between two specific cell means

Interpreting parameters

`lm()` gives **simple effects**

```
lm(formula = balance ~ hand * skill,  
    data = df.poker)
```

```
Call:
lm(formula = balance ~ hand * skill, data = df.poker)

Residuals:
    Min       1Q   Median       3Q      Max
-13.6976  -2.4740   0.0348   2.4644  14.7806

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.5866     0.5686   8.067 1.85e-14 ***
handneutral      5.2572     0.8041   6.538 2.75e-10 ***
handgood        -9.2110     0.8041  -11.455 < 2e-16 ***
skillexpert      2.7098     0.8041   3.370 0.000852 ***
handneutral:skillexpert -1.7042     1.1372  -1.499 0.133636
handgood:skillexpert  -4.2522     1.1372  -3.739 0.000222 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.02 on 294 degrees of freedom
Multiple R-squared:  0.3731,    Adjusted R-squared:  0.3624
F-statistic: 34.99 on 5 and 294 DF,  p-value: < 2.2e-16
```

`anova()` gives **main effects**,
and **interactions**

```
lm(formula = balance ~ hand * skill,  
    data = df.poker) %>%  
  anova()
```

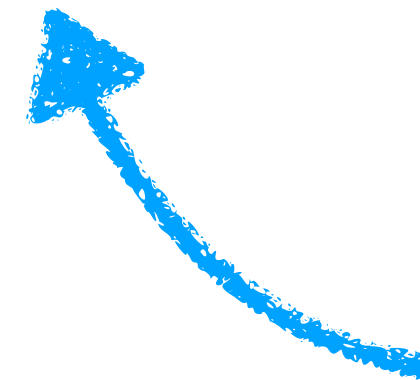
```
Analysis of Variance Table

Response: balance
          Df Sum Sq Mean Sq F value    Pr(>F)
hand       2 2559.4  1279.70   79.1692 < 2.2e-16 ***
skill      1   39.3    39.35    2.4344 0.1197776
hand:skill  2  229.0    114.49    7.0830 0.0009901 ***
Residuals 294 4752.3     16.16
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

anova () function

1. - model comparison: compare a compact with an augmented model to see whether the proportional reduction in error is worth it
2. - run an ANOVA to get main effects and interaction effects when you have categorical predictors as variables

Unbalanced designs



not the same number of
participants in each cell

ANOVA

- for all the examples so far, We've assumed a balanced design (i.e. the same number of observations in each of the different factor levels)
- things get *funky* when we have an unbalanced design



Beware of unbalanced designs

```
1 lm(formula = balance ~ skill + hand, data = df.poker.unbalanced) %>%
2   anova()
```

```
Analysis of Variance Table

Response: balance
          Df Sum Sq Mean Sq F value Pr(>F)
skill      1   74.3   74.28  4.2904 0.03922 *
hand       2 2385.1 1192.57 68.8827 < 2e-16 ***
Residuals 286 4951.5   17.31
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

flipped the order of the predictors

```
1 lm(formula = balance ~ hand + skill, data = df.poker.unbalanced) %>%
2   anova()
```

```
Analysis of Variance Table

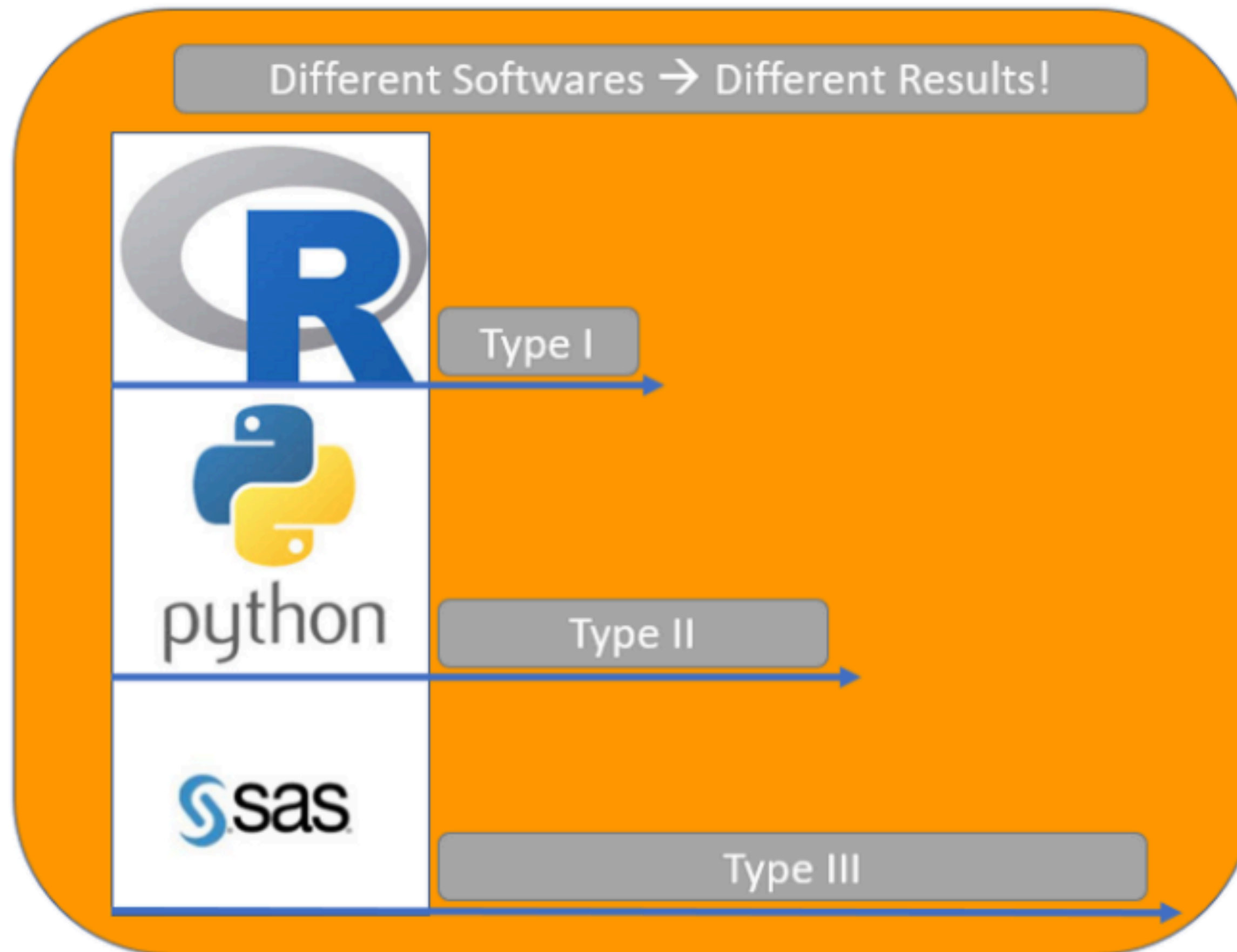
Response: balance
          Df Sum Sq Mean Sq F value Pr(>F)
hand       2 2419.8 1209.92 69.8845 <2e-16 ***
skill      1   39.6   39.59  2.2867 0.1316
Residuals 286 4951.5   17.31
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The different sums of squares

Three different methodologies for splitting variation exist: Type I, Type II and Type III Sums of Squares. They do not give the same result in case of unbalanced data.

Type I, Type II and Type III ANOVA have different outcomes!

Default sums of squares ...



Default Types of Sums of Squares for different programming languages

not great for reproducibility ...

Type I Sums of Squares

Type I Sums of Squares are Sequential, so the order of variables in the models makes a difference. This is rarely what we want in practice!

Sums of Squares are Mathematically defined as:

- $SS(A)$ for independent variable A
- $SS(B \mid A)$ for independent variable B
- $SS(AB \mid B, A)$ for the interaction effect

caution: this is what `anova()` uses by default

Type III Sums of Squares

The Type III Sums of Squares are also called partial sums of squares again another way of computing Sums of Squares:

- Like Type II, the Type III Sums of Squares are not sequential, so the order of specification does not matter.
- Unlike Type II, the Type III Sums of Squares do specify an interaction effect.

Sums of Squares are Mathematically defined as:

- $SS(A \mid B, AB)$ for independent variable A
- $SS(B \mid A, AB)$ for independent variable B

this is the default in the literature (e.g. SPSS, SAS, Stata etc use it)

Route 1: Using "afex"

```
1 library("afex")
2
3 fit = aov_ez(id = "participant",
4             dv = "balance",
5             data = df.poker.unbalanced,
6             between = c("hand", "skill"))
7 fit$Anova
```

Contrasts set to contr.sum for the following variables: hand, skill

Anova Table (Type III tests)

Response: dv

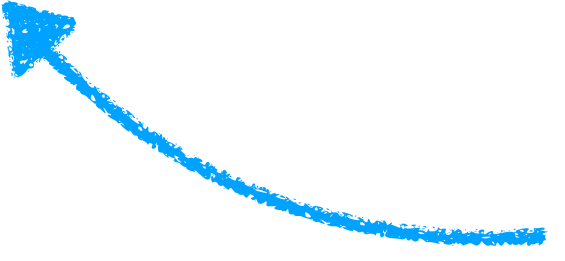
	Sum Sq	Df	F value	Pr(>F)	
(Intercept)	27781.3	1	1676.9096	< 2.2e-16	***
hand	2285.3	2	68.9729	< 2.2e-16	***
skill	48.9	1	2.9540	0.0867525	.
hand:skill	246.5	2	7.4401	0.0007089	***
Residuals	4705.0	284			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Route II: Using "emmeans"

**preferred
route!!**

```
1 library("emmeans")
2
3 lm(formula = balance ~ hand * skill,
4     data = df.poker.unbalanced) %>%
5     joint_tests()
```

 **very handy function**

model	term	df1	df2	F.ratio	p.value
	hand	2	284	68.973	<.0001
	skill	1	284	2.954	0.0868
	hand:skill	2	284	7.440	0.0007

Unbalanced design

- There are different kinds of ANOVAs, for which the sums of squares are calculated differently.
- This makes a difference when we have an unbalanced design (i.e. the number of participants is not the same for each cell in our design).

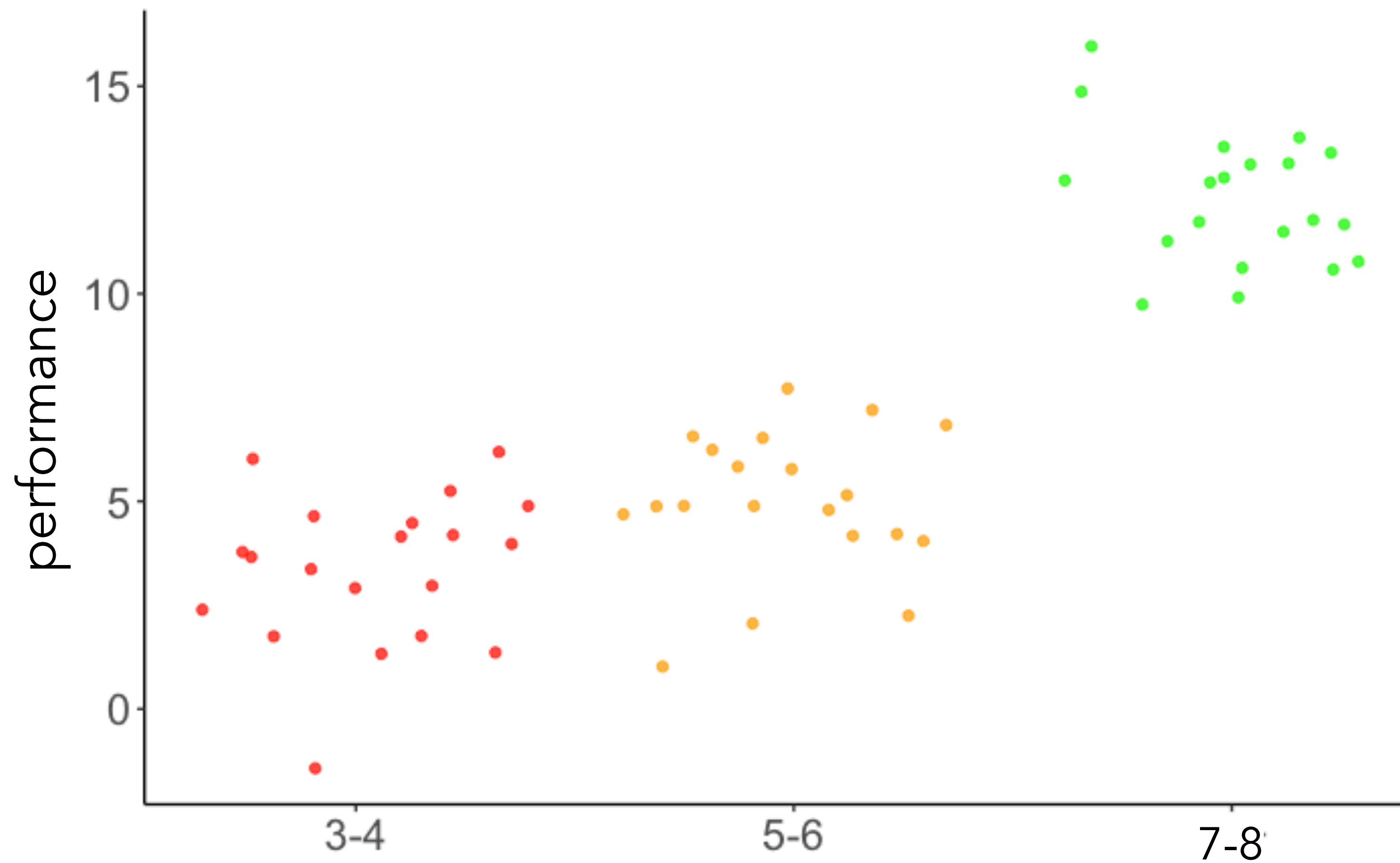
`joint_tests()` is your friend!

Linear contrasts

Testing (more) specific hypotheses with linear contrasts

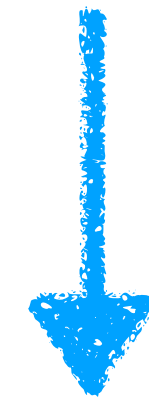
Contrasts

Does performance increase with age?



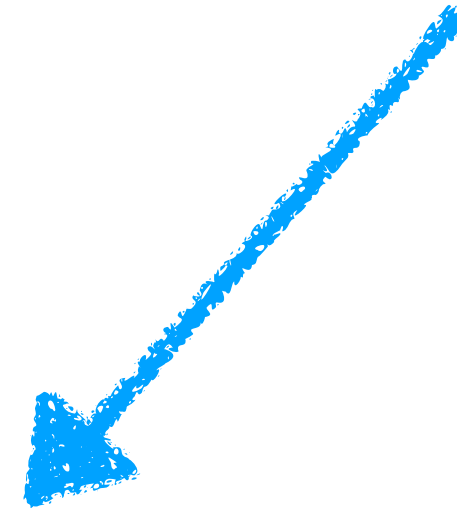
Data from a hypothetical developmental study

Does performance increase with age?



ANOVA

Does performance differ between age groups?



3-4 vs. 5-6

post-hoc tests



5-6 vs. 7-8



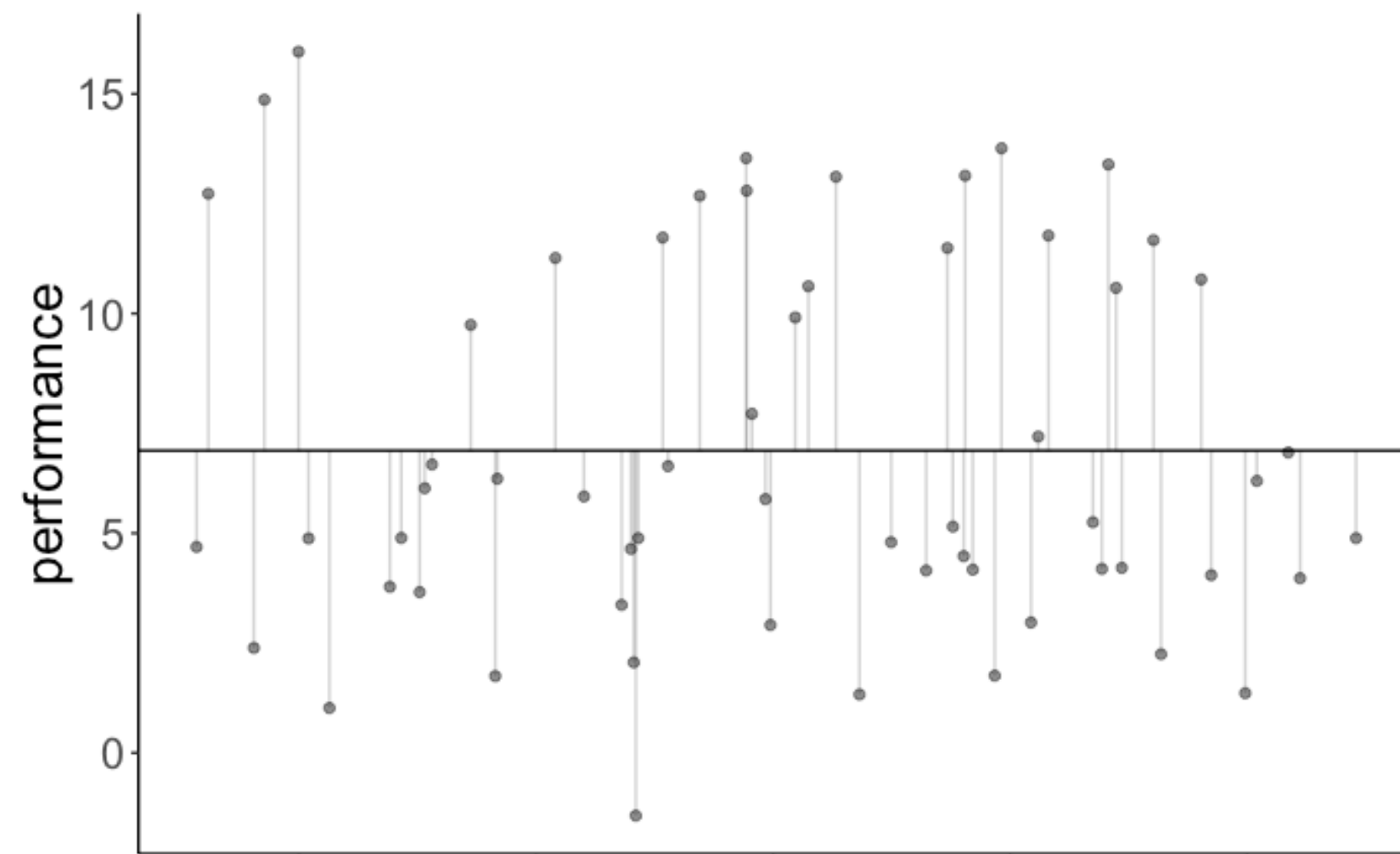
Is there are more direct way of asking this question with a statistical model?

Contrasts

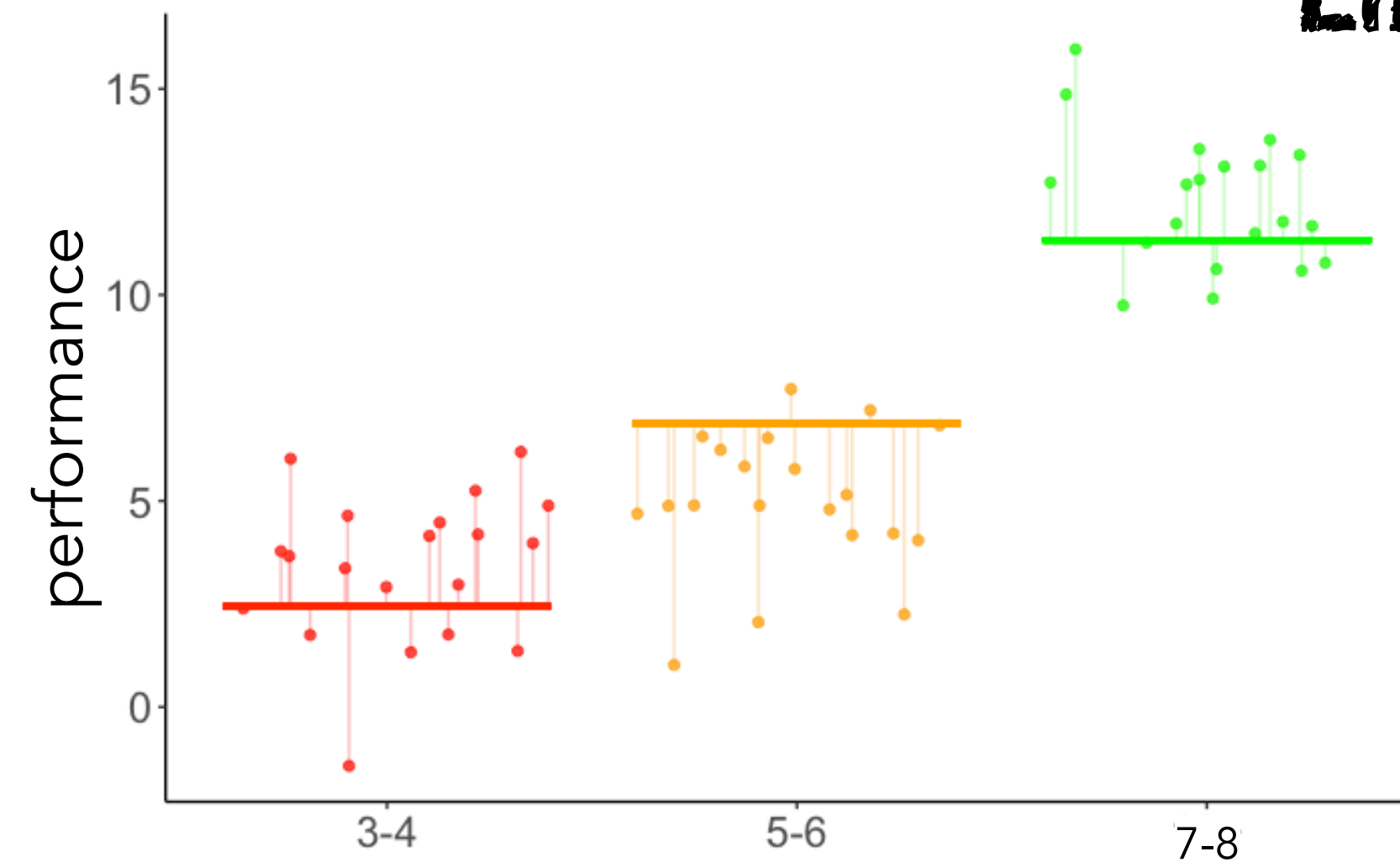
Does performance increase with age?

`contrasts = c(-1, 0, 1)`

Compact model



Augmented model



Linear contrast

Model comparison

$p < .001$

`emmeans` **for handling linear contrasts in R**

Linear contrasts

~~How to use contrasts in R~~

In short: don't bother.¹

Like many before me, one of my stats classes technically “taught” me contrasts. But I didn't get the point and using them was cumbersome, so I promptly ignored them for years.

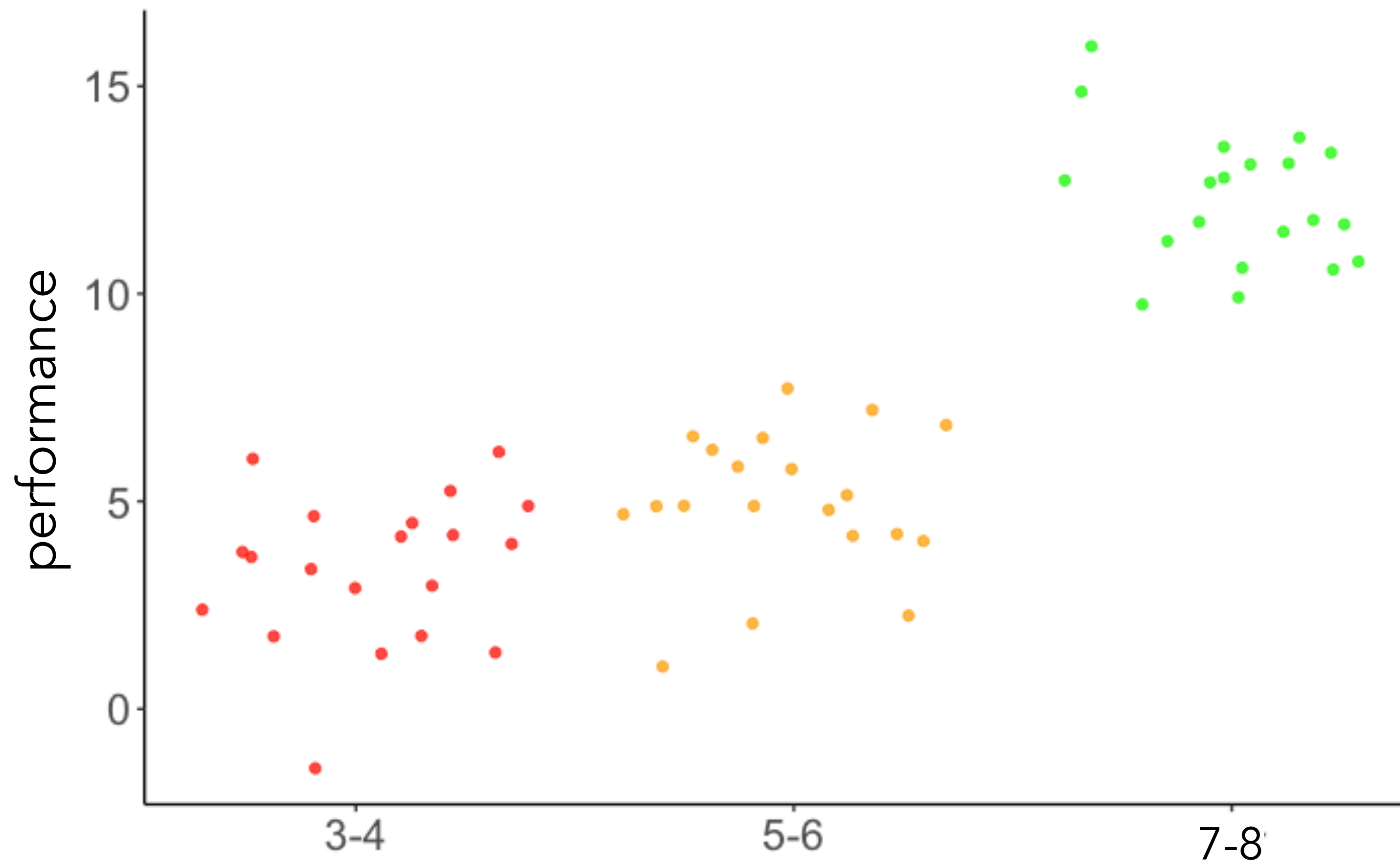
Luckily for me, someone came along and fixed the situation: [emmeans](#). `emmeans` frames contrasts as a question you pose to a model: you can ask for all pairwise comparisons and get back that. `lm` and `summary` treat the same problem as fitting abstract coefficients, and you are left to answer your own question.

`emmeans` works with `lm`, `glm`, and the Bayesian friends in [brms](#) and [rstanarm](#), so the process is applicable no matter the tool.

And you don't have to learn (much) about contrasts to take advantage of it.

Contrasts

Does performance increase with age?



Data from a hypothetical developmental study

Linear contrasts in R

```
1 library("emmeans") # for calculating contrasts
2
3 # fit the linear model
4 fit = lm(formula = performance ~ 1 + group,
5           data = df.development)
```

fit linear model

Linear contrasts in R

```
1 library("emmeans") # for calculating contrasts
2
3 # fit the linear model
4 fit = lm(formula = performance ~ 1 + group,
5          data = df.development)
6
7 # check factor levels
8 levels(df.development$group)      [1] "3-4" "5-6" "7-8"
```

**check factor levels before
defining contrasts**

Linear contrasts in R

```
1 library("emmeans") # for calculating contrasts
2
3 # fit the linear model
4 fit = lm(formula = performance ~ group,
5          data = df.development)
6
7 # check factor levels
8 levels(df.development$group) [1] "3-4" "5-6" "7-8"
9
10 # define the contrasts of interest
11 contrasts = list(young_vs_old = c(-0.5, -0.5, 1),
12                 three_vs_five = c(-0.5, 0.5, 0))
```

set up linear contrasts

Linear contrasts in R

```
1 library("emmeans") # for calculating contrasts
2
3 # fit the linear model
4 fit = lm(formula = performance ~ group,
5          data = df.development)
6
7 # check factor levels
8 levels(df.development$group) [1] "3-4" "5-6" "7-8"
9
10 # define the contrasts of interest
11 contrasts = list(young_vs_old = c(-0.5, -0.5, 1),
12                 three_vs_five = c(-0.5, 0.5, 0))
13
14 # compute significance test on contrasts
15 fit %>%
16   emmeans("group",
17          contr = contrasts,
18          adjust = "bonferroni") %>%
19   pluck("contrasts")
```

compute the results

```
[1] "3-4" "5-6" "7-8"
contrast      estimate      SE df t.ratio p.value
young_vs_old  16.093541 0.4742322 57  33.936 <.0001
three_vs_five  1.606009 0.5475962 57   2.933  0.0097

P value adjustment: bonferroni method for 2 tests
```

Linear contrasts in R

```
1 library("emmeans") # for calculating contrasts
2
3 # fit the linear model
4 fit = lm(formula = performance ~ group,
5          data = df.development)
6
7 # check factor levels
8 levels(df.development$group) [1] "3-4" "5-6" "7-8"
9
10 # define the contrasts of interest
11 contrasts = list(young_vs_old = c(-1, -1, 2),
12                 three_vs_five = c(-1, 1, 0))
13
14 # compute significance test on contrasts
15 fit %>%
16   emmeans("group",
17           contr = contrasts,
18           adjust = "bonferroni") %>%
19   pluck("contrasts")
```

hypothesis tests
are the same!

```
# define the contrasts of interest
contrasts = list(young_vs_old = c(-0.5, -0.5, 1),
                 three_vs_five = c(-0.5, 0.5, 0))
```

	contrast	estimate	SE	df	t.ratio	p.value
[1] "3-4" "5-6" "7-8"	young_vs_old	16.093541	0.4742322	57	33.936	<.0001
	three_vs_five	1.606009	0.5475962	57	2.933	0.0097

P value adjustment: bonferroni method for 2 tests

	contrast	estimate	SE	df	t.ratio	p.value
[1] "3-4" "5-6" "7-8"	young_vs_old	32.187	0.948	57	33.936	<.0001
	three_vs_five	0.803	0.274	57	2.933	0.0097

P value adjustment: bonferroni method for 2 tests

Defining contrasts

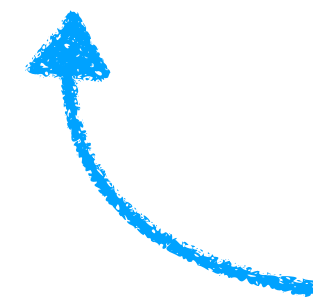
- groups that we don't want to include in the comparison get a 0
- groups that we want to compare with one another should sum to 0
- Thus, all the contrasts together should sum to 0

Example:

```
contrasts = list(young_vs_old = c(-1, -1, 2),  
                 three_vs_five = c(-1, 1, 0))
```

Post hoc tests

```
1 fit = lm(formula = performance ~ 1 + group,  
2          data = df.development)  
3  
4 # pairwise differences between all the groups  
5 fit %>%  
6   emmeans(pairwise ~ group) %>%  
7   pluck("contrasts")
```



all pairwise tests between groups

contrast	estimate	SE	df	t.ratio	p.value
3-4 - 5-6	-1.606009	0.5475962	57	-2.933	0.0145
3-4 - 7-8	-16.896546	0.5475962	57	-30.856	<.0001
5-6 - 7-8	-15.290537	0.5475962	57	-27.923	<.0001

P value adjustment: bonferroni method for 3 tests

Post hoc tests

```
1 # fit the model
2 fit = lm(formula = balance ~ 1 + hand + skill,
3          data = df.poker)
4
5 # post hoc tests
6 fit %>%
7   emmeans(pairwise ~ hand + skill,
8           adjust = "bonferroni") %>%
9   pluck("contrasts")
```

← the poker data

contrast	estimate	SE	df	t.ratio	p.value
bad,average - neutral,average	-4.381023	0.6051766	286	-7.239	<.0001
bad,average - good,average	-7.060823	0.6051766	286	-11.667	<.0001
bad,average - bad,expert	-0.740385	0.4896119	286	-1.512	1.0000
bad,average - neutral,expert	-5.121408	0.7611327	286	-6.729	<.0001
bad,average - good,expert	-7.801208	0.7611327	286	-10.249	<.0001
neutral,average - good,average	-2.679800	0.5884403	286	-4.554	0.0001
neutral,average - bad,expert	3.640638	0.7953578	286	4.577	0.0001
neutral,average - neutral,expert	-0.740385	0.4896119	286	-1.512	1.0000
neutral,average - good,expert	-3.420185	0.7654945	286	-4.468	0.0002
good,average - bad,expert	6.320438	0.7953578	286	7.947	<.0001
good,average - neutral,expert	1.939415	0.7654945	286	2.534	0.1774
good,average - good,expert	-0.740385	0.4896119	286	-1.512	1.0000
bad,expert - neutral,expert	-4.381023	0.6051766	286	-7.239	<.0001
bad,expert - good,expert	-7.060823	0.6051766	286	-11.667	<.0001
neutral,expert - good,expert	-2.679800	0.5884403	286	-4.554	0.0001

P value adjustment: bonferroni method for 15 tests

that's a lot of tests!

... not confirmatory

all pairwise tests between groups

Contrasts

- linear contrasts allow us to ask more specific questions of our data
- rather than asking whether any of the group means are significantly different from each other (ANOVA), we can ask questions such as:
 - Does performance differ with age?
 - Is the overall performance in Condition B and C better from the performance in Condition A?

Generalized linear model

Titanic dataset



Titanic data set

891 passengers

passenger_id	survived	pclass	name	sex	age	sib_sp	parch	ticket	fare	cabin	embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs	female	38	1	0	PC 17599	71.28	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.92		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.10	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male	NA	0	0	330877	8.46		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.86	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.07		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth	female	27	0	2	347742	11.13		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.07		C

Is there a relationship between fare and survived?

```
1 fit.lm = lm(formula = survived ~ 1 + fare,  
2           data = df.titanic)  
3  
4 fit.lm %>% summary()
```

Call:

```
lm(formula = survived ~ 1 + fare, data = df.titanic)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.9653	-0.3391	-0.3222	0.6044	0.6973

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.3026994	0.0187849	16.114	< 2e-16 ***
fare	0.0025195	0.0003174	7.939	6.12e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

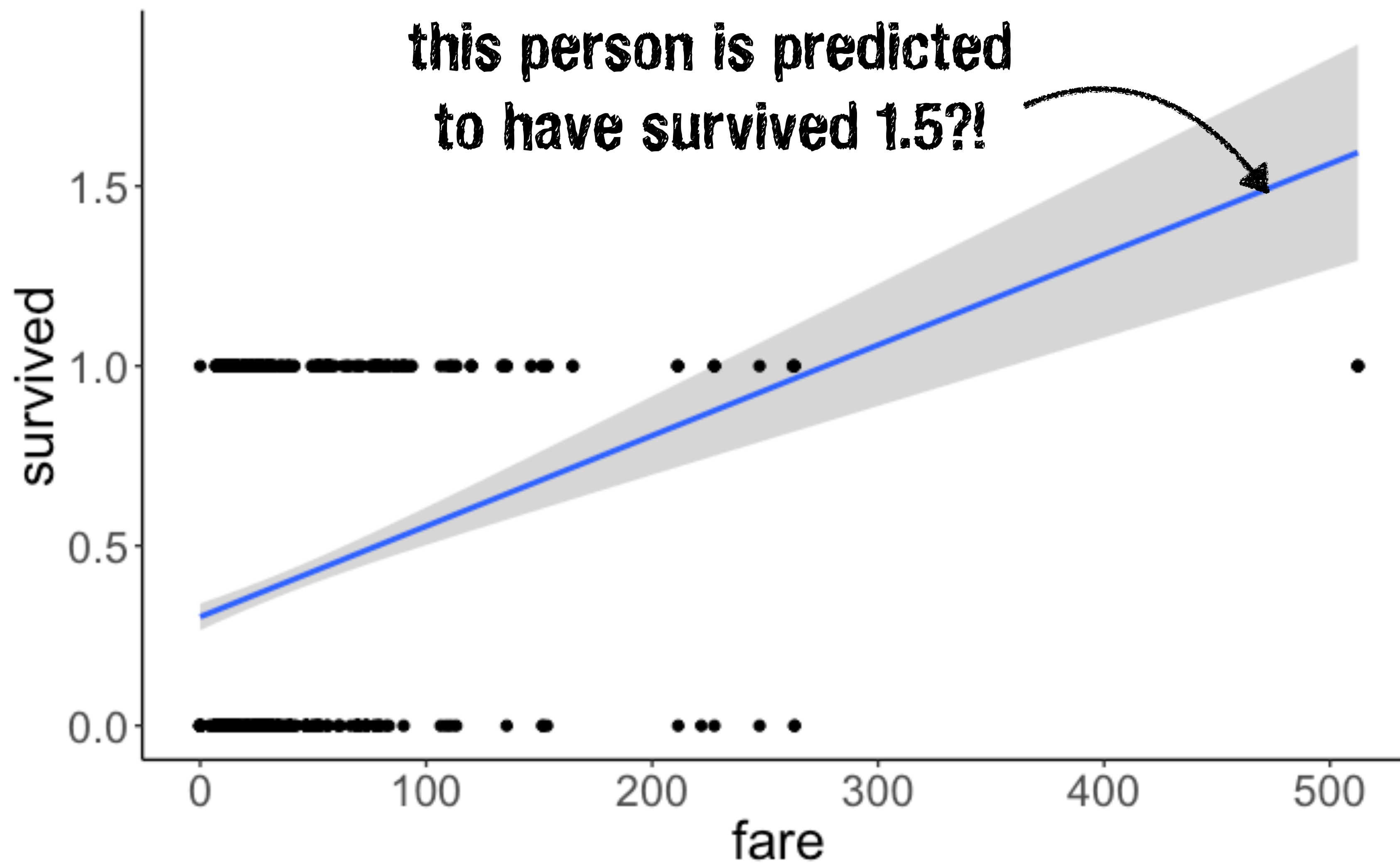
Residual standard error: 0.4705 on 889 degrees of freedom

Multiple R-squared: 0.06621, Adjusted R-squared: 0.06516

F-statistic: 63.03 on 1 and 889 DF, p-value: 6.12e-15

**How should we
interpret this
parameter?**

Is there a relationship between `fare` and `survived`?



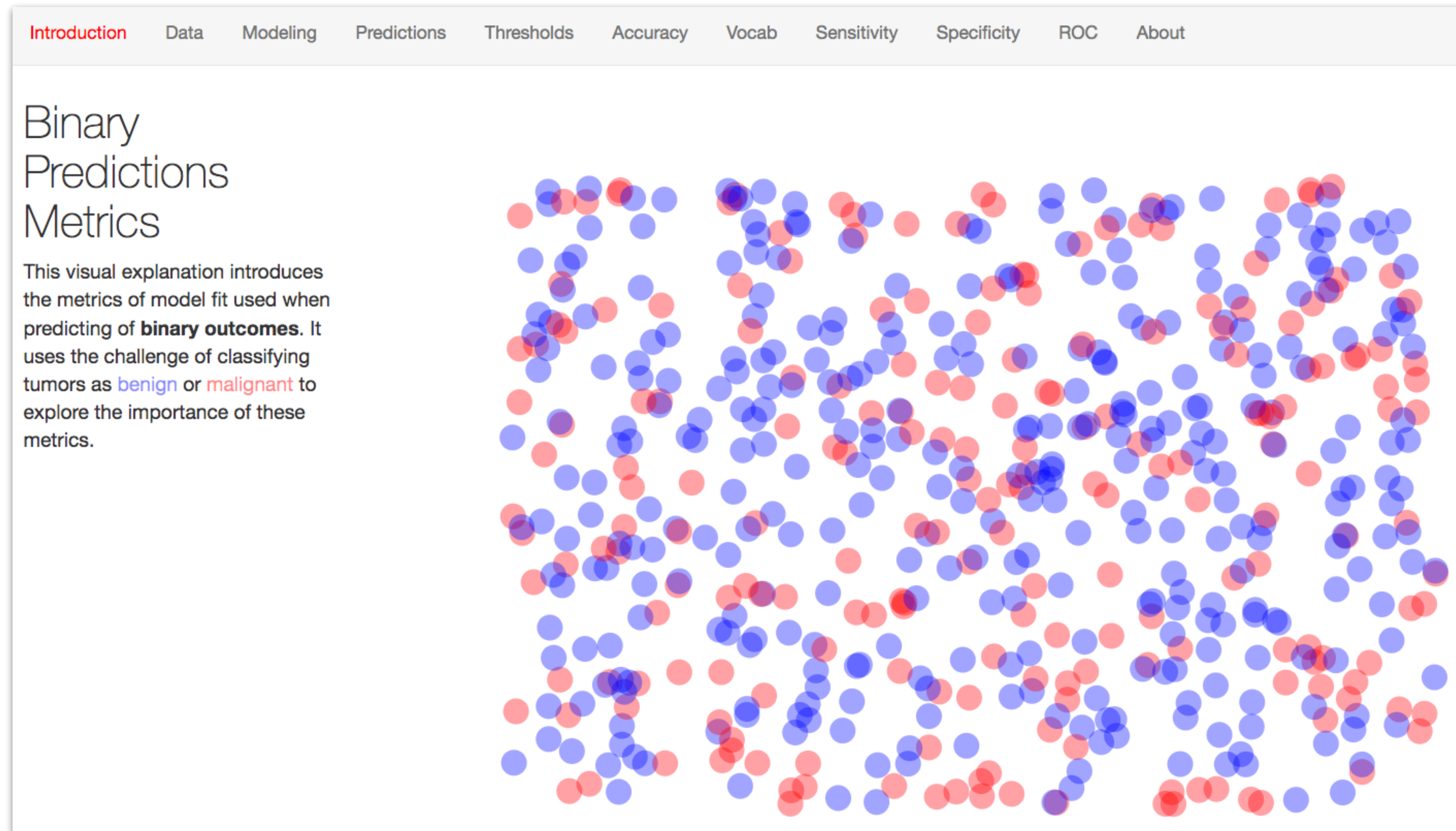
Generalized linear model

- so far, we have only looked at situations where our outcome variable was **continuous**
- what about situations in which we have a **binary** outcome variable?
 - survived vs. died
 - correct vs. incorrect
 - benign vs. malignant
 - yes vs. no
 - ...



Logistic regression

Demo



<http://mfviz.com/binary-predictions/>

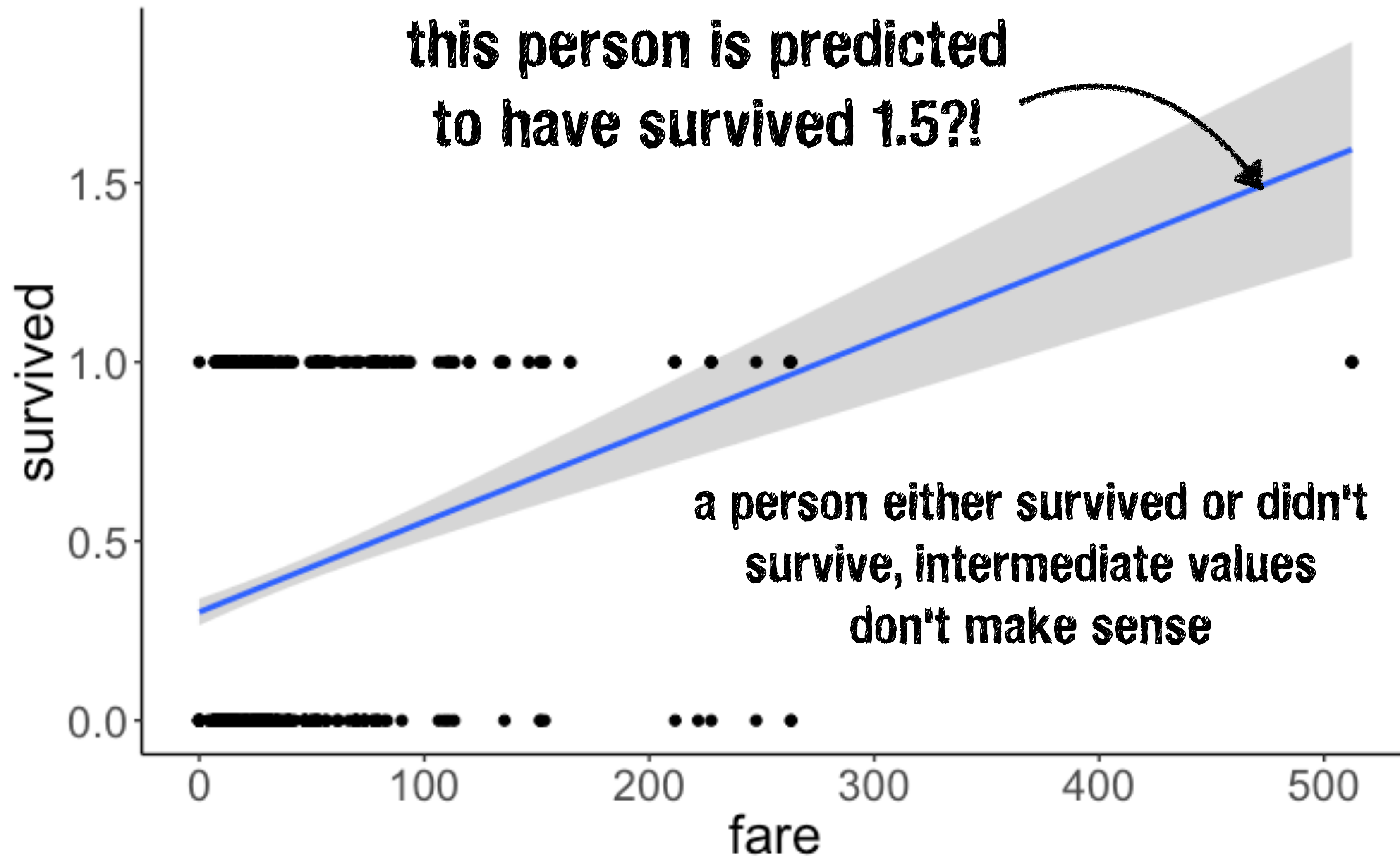
Is there a relationship between `fare` and `survived`?

**Can we still use a linear model to
make predictions about a binary
outcome variable?**

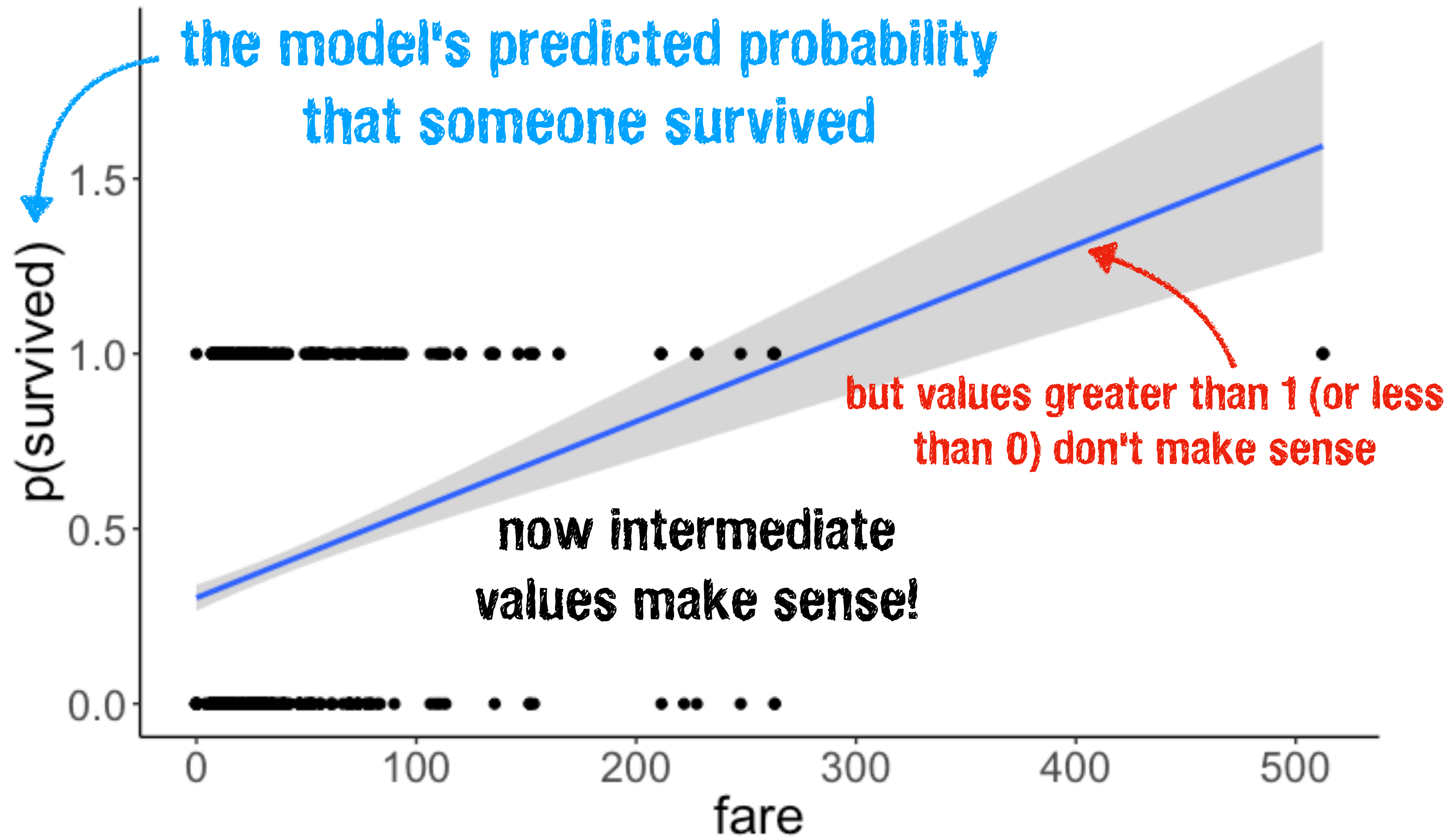
The fact that this class is
called "**Generalized linear
model**" suggests we can!

Is there a relationship between fare and survived?

```
fit.lm = lm(formula = survived ~ 1 + fare, data = df.titanic)
```



Is there a relationship between fare and survived?



From linear regression to logistic regression

$$Y_i = b_0 + b_1 \cdot X_i + e_i$$

predict the value of Y $[-\infty, +\infty]$

$$\pi_i = \frac{e^{(b_0 + b_1 \cdot X_i)}}{1 + e^{(b_0 + b_1 \cdot X_i)}}$$

predict the probability of Y $[0, 1]$

$$\pi_i = P(Y_i = 1)$$

let's just do a
logit transform

we need to map between $[-\infty, +\infty]$ to $[0, 1]$

Logit transform

predict the probability of Y

$$\pi_i = P(Y_i = 1)$$

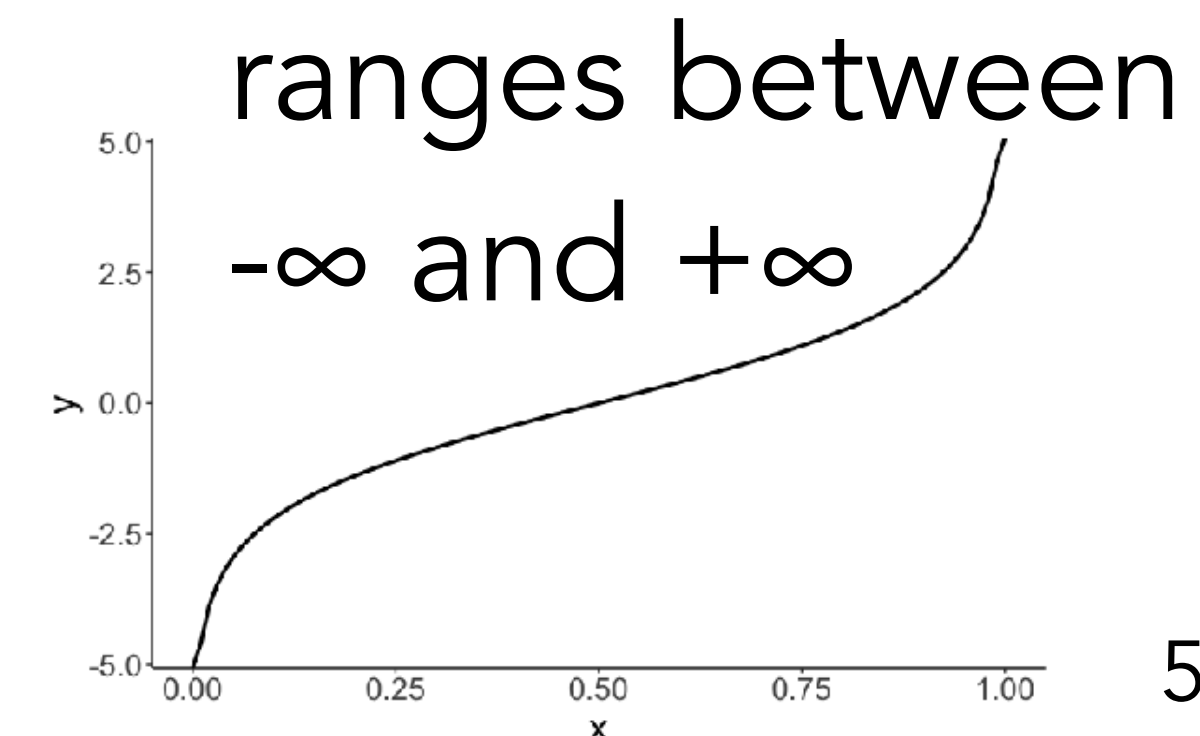
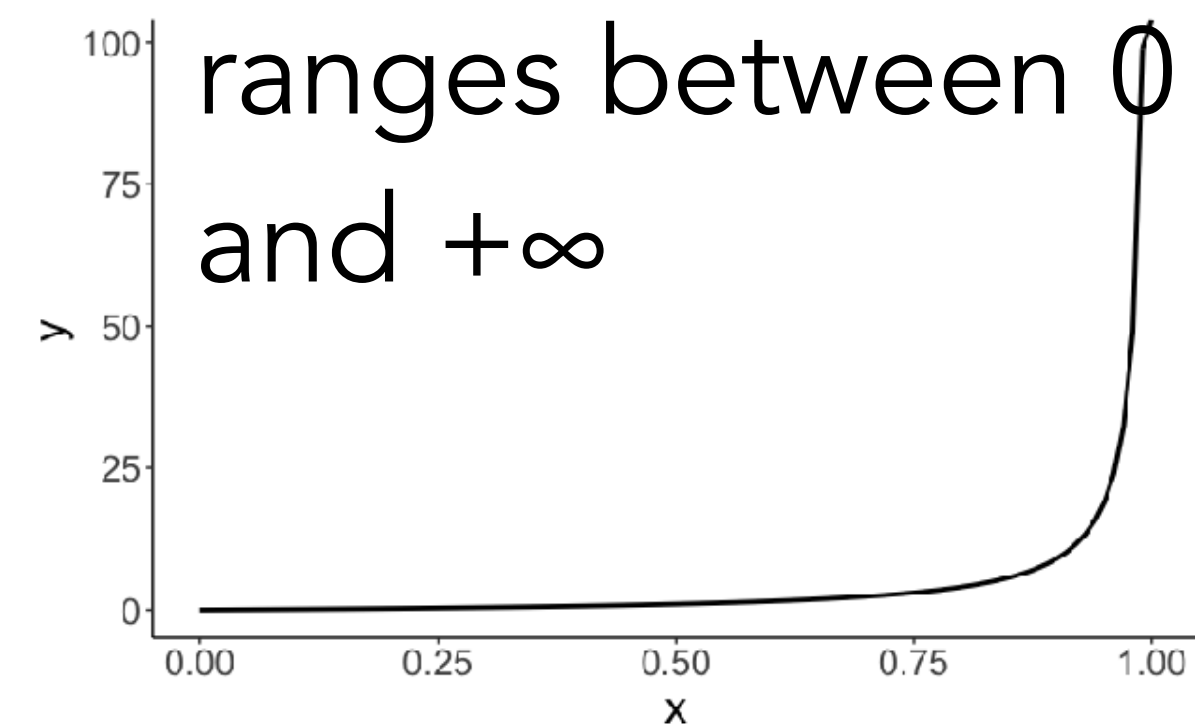
we need to transform the dependent variable so that it can take any value between $-\infty$ and $+\infty$ (we can then transform it back into a probability later)

Step 1: Calculate the "odds"

$$\frac{P(Y_i = 1)}{P(Y_i = 0)} = \frac{\pi_i}{1 - \pi_i}$$

Step 2: Take the (natural) log

$$\ln\left(\frac{\pi_i}{1 - \pi_i}\right) = b_0 + b_1 \cdot X_i$$



Logit transform

log odds

after transforming from a binary variable, to a probability, to odds, to log odds, the model looks like a normal linear model

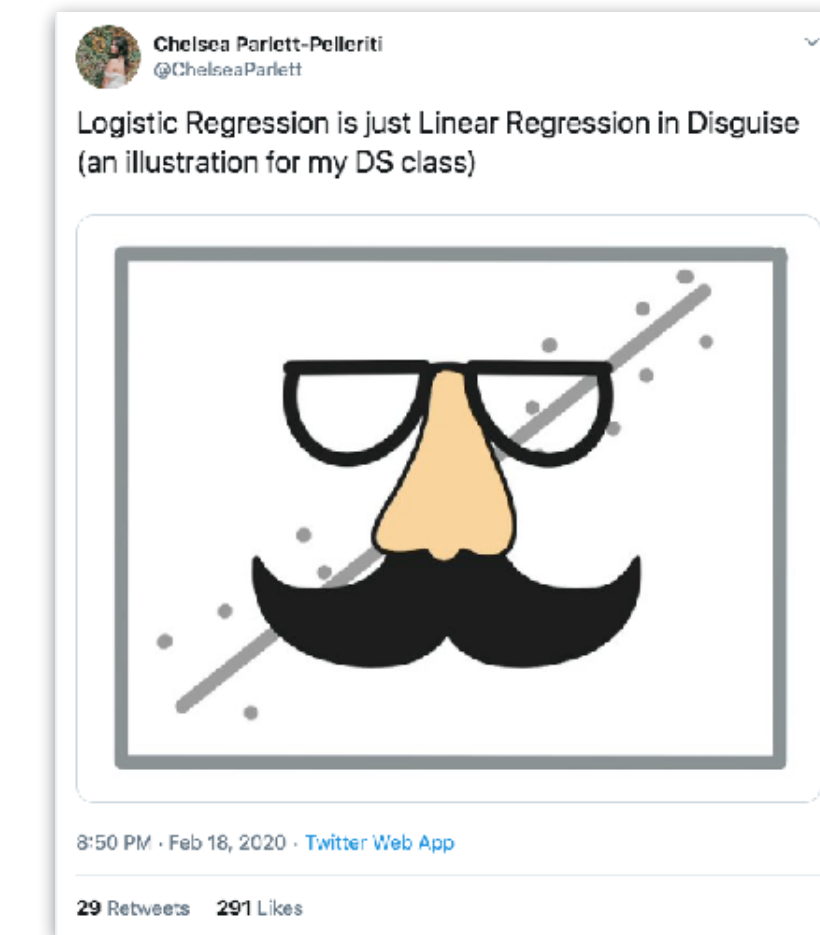
$$\ln\left(\frac{\pi_i}{1 - \pi_i}\right) = b_0 + b_1 \cdot X_i$$

$$\pi_i = P(Y_i = 1)$$

if log odds == 0: $P(Y_i = 1) = P(Y_i = 0)$

if log odds > 0: $P(Y_i = 1) > P(Y_i = 0)$

if log odds < 0: $P(Y_i = 1) < P(Y_i = 0)$



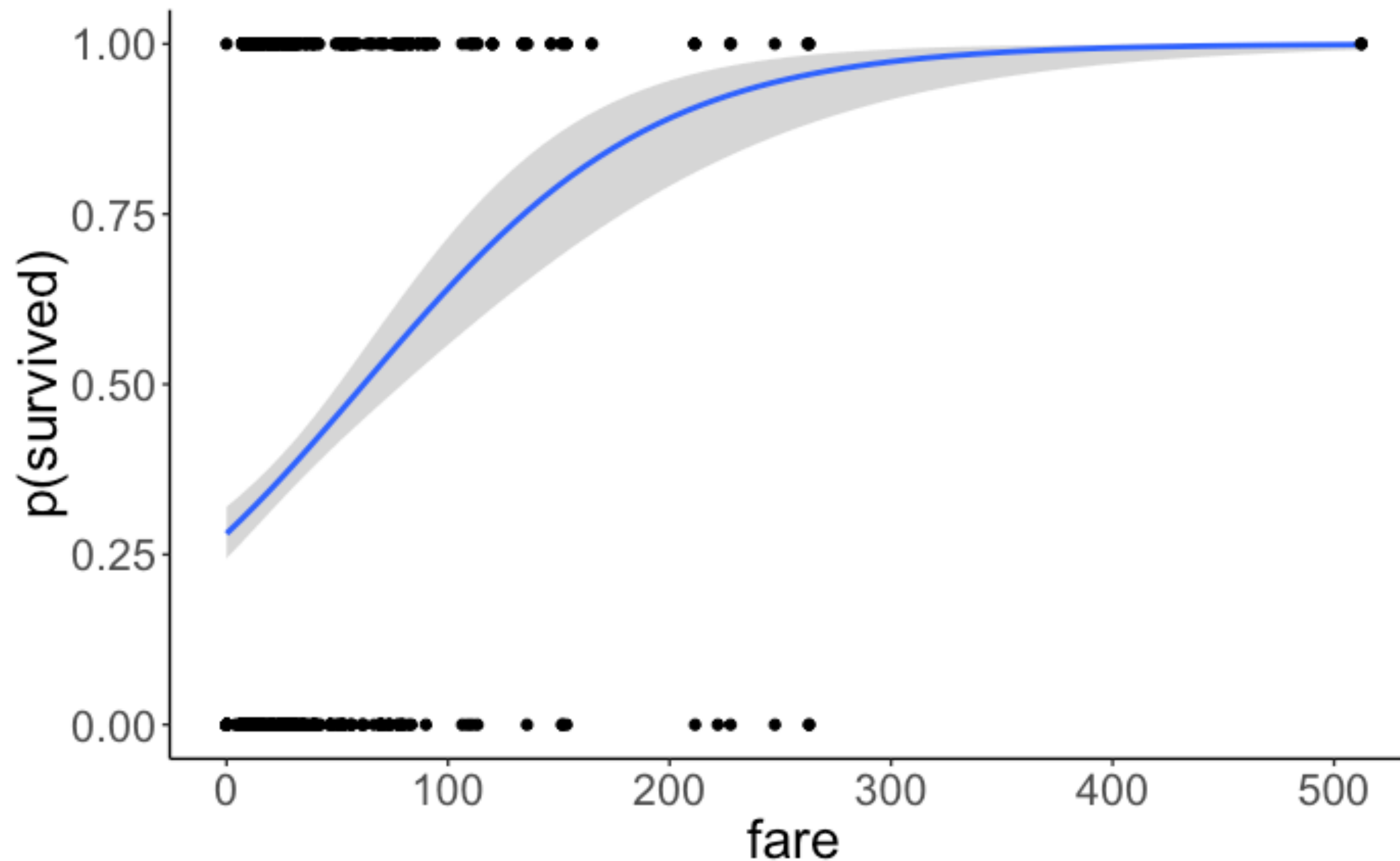
Fitting a logistic regression in R

```
1 fit.glm = glm(formula = survived ~ 1 + fare,  
2               family = "binomial",  
3               data = df.titanic)  
4  
5 fit.glm %>% summary()
```

```
Call:  
glm(formula = survived ~ 1 + fare, family = "binomial", data = df.titanic)  
  
Deviance Residuals:  
    Min       1Q   Median       3Q      Max   
-2.4906  -0.8878  -0.8531   1.3429   1.5942   
  
Coefficients:  
              Estimate Std. Error z value Pr(>|z|)      
(Intercept) -0.941330   0.095129  -9.895  < 2e-16 ***  
fare         0.015197   0.002232   6.810 9.79e-12 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
(Dispersion parameter for binomial family taken to be 1)  
  
    Null deviance: 1186.7  on 890  degrees of freedom  
Residual deviance: 1117.6  on 889  degrees of freedom  
AIC: 1121.6  
  
Number of Fisher Scoring iterations: 4
```

Visualize the model's predictions

```
1 ggplot(data = df.titanic,  
2         mapping = aes(x = fare,  
3                       y = survived)) +  
4   geom_smooth(method = "glm",  
5              method.args = list(family = "binomial")) +  
6   geom_point() +  
7   labs(y = "p(survived)")
```



Interpreting the model output

Interpreting the model output

```
Call:
glm(formula = survived ~ 1 + fare, fa:

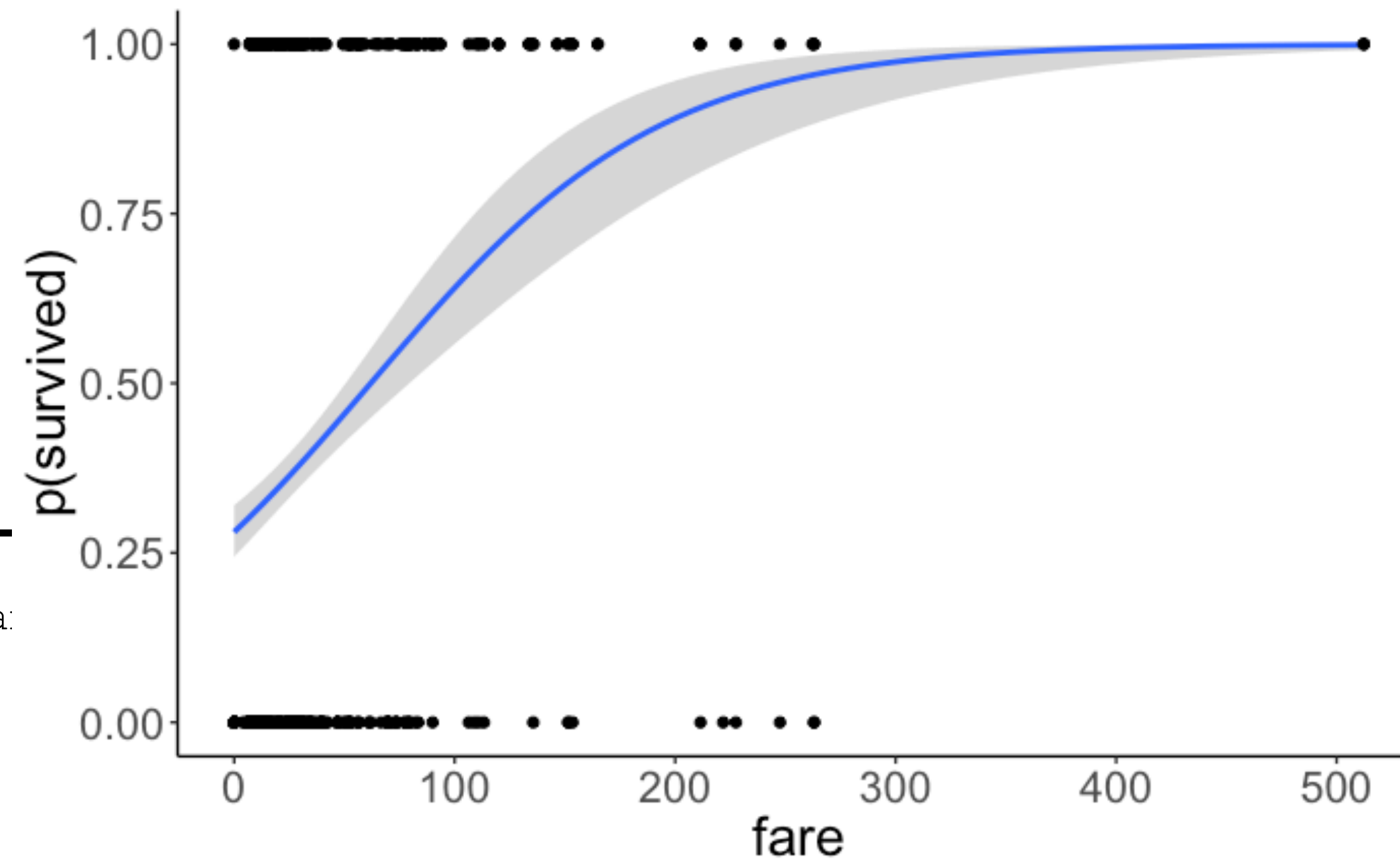
Deviance Residuals:
    Min       1Q   Median       3Q      Max 
-2.4006  -0.8878  -0.8538   1.3429   2.4006 

log odds ?
Coefficients:
(Intercept) -0.941330 0.095129 -9.895 < 2e-16 ***
fare         0.015197 0.002232  6.810 9.79e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1186.7  on 890  degrees of freedom
Residual deviance: 1117.6  on 889  degrees of freedom
AIC: 1121.6

Number of Fisher Scoring iterations: 4
```



Transform log odds into probability

$$\pi = P(Y = 1)$$

$$\ln\left(\frac{\pi}{1 - \pi}\right) = V$$

just a placeholder

logit transformation

$$\pi = \frac{e^V}{1 + e^V}$$

inverse logit

gives us back the probability
(which is much easier to interpret)

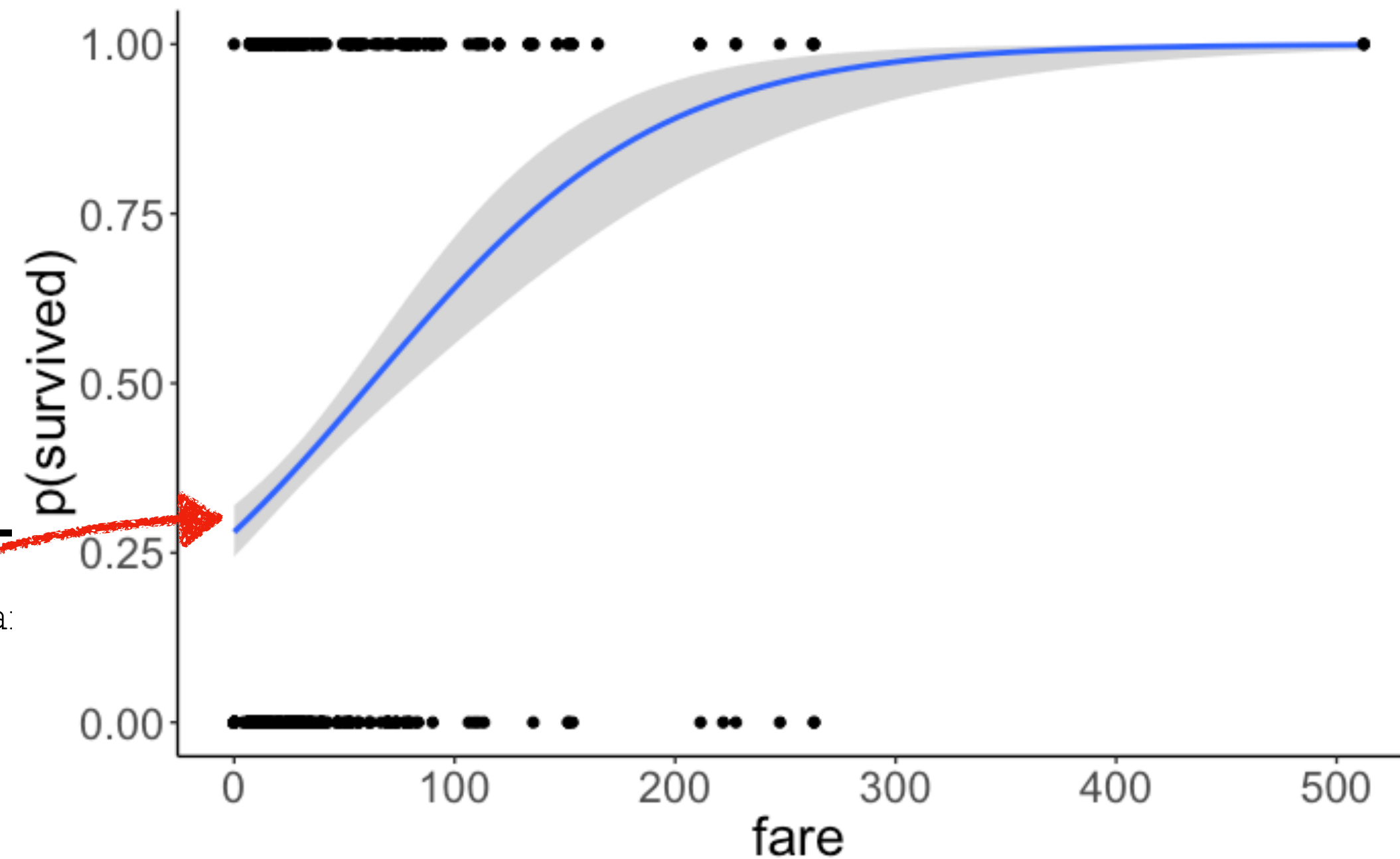
$$\pi_i = \frac{e^{b_0 + b_1 \cdot X_i}}{1 + e^{b_0 + b_1 \cdot X_i}}$$

another way to
specify the model

Interpreting the model output

inverse logit

$$\pi = \frac{e^{-0.94}}{1 + e^{-0.94}} \approx 0.28$$



```
Call:
glm(formula = survived ~ 1 + fare, fa:

Deviance Residuals:
    Min       1Q   Median       3Q      Max 
-2.4906  -0.8878  -0.8531   1.3429   2.4906 

Coefficients:
(Intercept) -0.941330 
fare         0.015197 

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1186.7  on 890  degrees of freedom
Residual deviance: 1117.6  on 889  degrees of freedom
AIC: 1121.6

Number of Fisher Scoring iterations: 4
```

Interpreting the model output

```
Call:
glm(formula = survived ~ 1 + fare, fa:

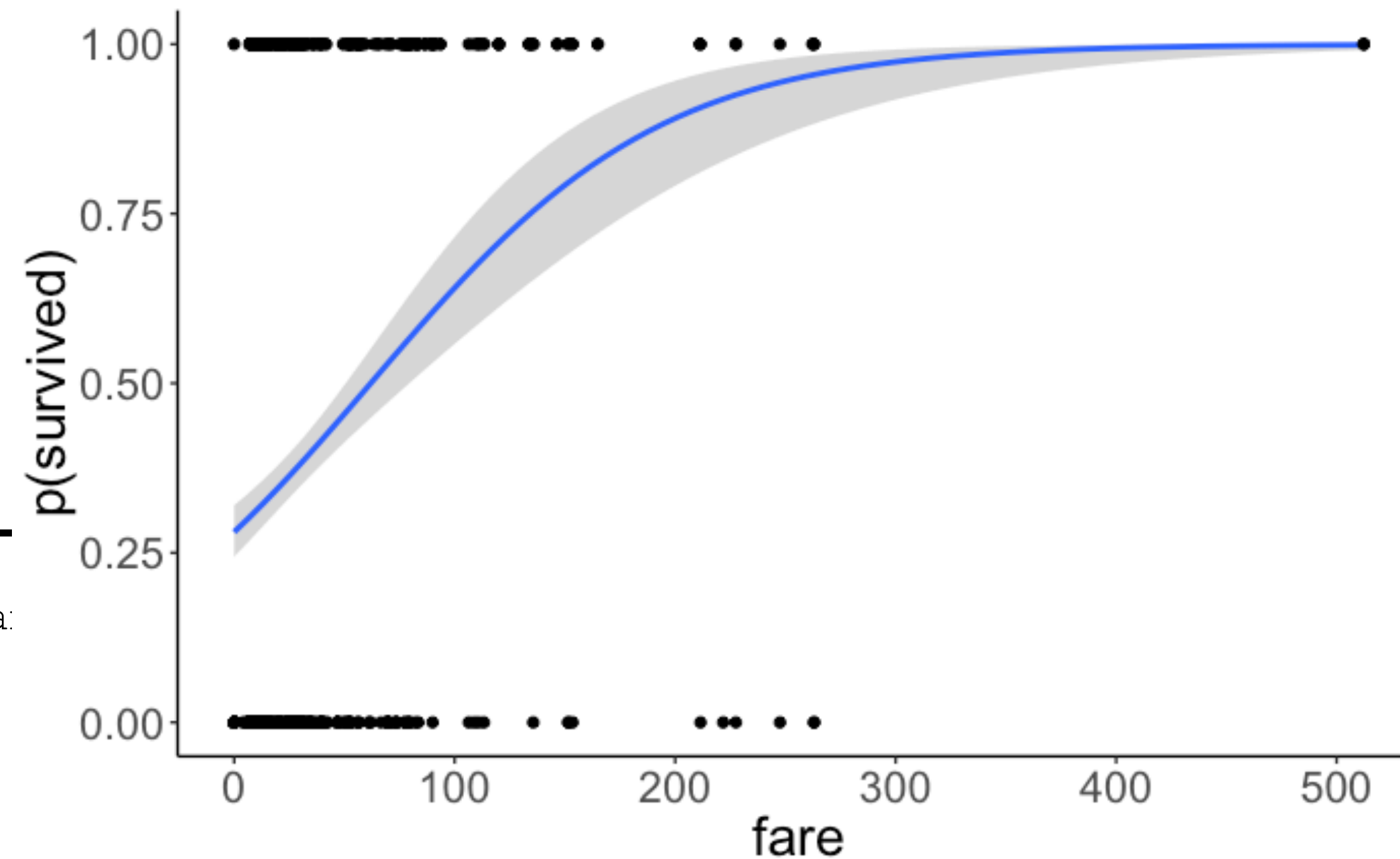
Deviance Residuals:
    Min       1Q   Median       3Q      Max 
-2.4906  -0.8878  -0.8521   1.3429   2.4906 

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.941330   0.095129  -9.895  < 2e-16 ***
fare         0.015197   0.002232   6.810  9.79e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

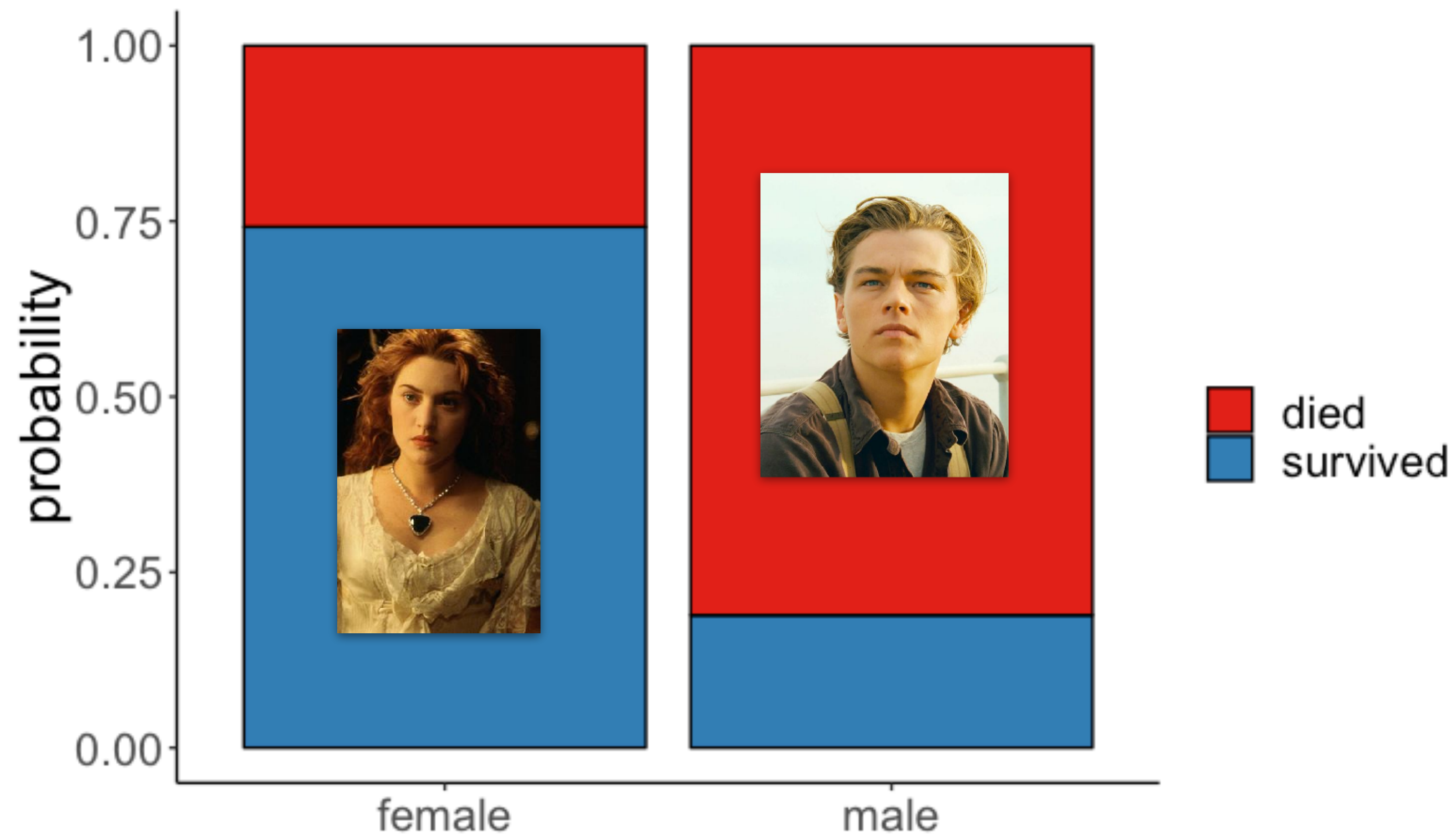
    Null deviance: 1186.7  on 890  degrees of freedom
Residual deviance: 1117.6  on 889  degrees of freedom
AIC: 1121.6

Number of Fisher Scoring iterations: 4
```



Let's consider a binary predictor

Was the probability of survival different between female and male passengers on the Titanic?



Let's consider a binary predictor

```
1 fit.glm2 = glm(formula = survived ~ 1 + sex,  
2               family = "binomial",  
3               data = df.titanic)  
4  
5 fit.glm2 %>% summary()
```

```
Call:  
glm(formula = survived ~ sex, family = "binomial", data = df.titanic)
```

```
Deviance Residuals:
```

Min	1Q	Median	3Q	Max
-1.6462	-0.6471	-0.6471	0.7725	1.8256

```
Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.0566	0.1290	8.191	2.58e-16 ***
sexmale	-2.5137	0.1672	-15.036	< 2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 1186.7 on 890 degrees of freedom  
Residual deviance: 917.8 on 889 degrees of freedom  
AIC: 921.8
```

```
Number of Fisher Scoring iterations: 4
```

**sex was significantly
associated with survival**

Let's consider a binary predictor

$$\ln\left(\frac{p(\text{survived})_i}{1 - p(\text{survived})_i}\right) = b_0 + b_1 \cdot \text{sex}_i$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.0566	0.1290	8.191	2.58e-16	***
sexmale	-2.5137	0.1672	-15.036	< 2e-16	***

sex	survived	n	p	p(survived sex)
female	0	81	0.09	0.26
female	1	233	0.26	0.74
male	0	468	0.53	0.81
male	1	109	0.12	0.19

if sex == 0:

$$\ln\left(\frac{\widehat{p(\text{survived})}_i}{1 - \widehat{p(\text{survived})}_i}\right) = b_0$$

$$p(\text{survived})_i = \frac{e^{b_0}}{1 + e^{b_0}} = 0.74$$

Let's consider a binary predictor

$$\ln\left(\frac{p(\text{survived})_i}{1 - p(\text{survived})_i}\right) = b_0 + b_1 \cdot \text{sex}_i$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.0566	0.1290	8.191	2.58e-16	***
sexmale	-2.5137	0.1672	-15.036	< 2e-16	***

sex	survived	n	p	p(survived sex)
female	0	81	0.09	0.26
female	1	233	0.26	0.74
male	0	468	0.53	0.81
male	1	109	0.12	0.19

if sex == 1:

$$\ln\left(\frac{\widehat{p(\text{survived})}_i}{1 - \widehat{p(\text{survived})}_i}\right) = b_0 + b_1$$

$$p(\text{survived})_i = \frac{e^{b_0+b_1}}{1 + e^{b_0+b_1}} = 0.19$$

Now let's go back to a continuous predictor

$$\ln\left(\frac{p(\text{survived})_i}{1 - p(\text{survived})_i}\right) = b_0 + b_1 \cdot \text{fare}_i$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.941330	0.095129	-9.895	< 2e-16	***
fare	0.015197	0.002232	6.810	9.79e-12	***

fare	prediction	p(survival)
0	-0.94	0.28
10	-0.79	0.31
50	-0.18	0.45
100	0.58	0.64
500	6.66	1.00

$$\ln\left(\frac{\widehat{p(\text{survived})}}{1 - \widehat{p(\text{survived})}}\right) = -0.94 + 0.015 \cdot 10$$

$$p(\text{survived})_i = \frac{e^{-0.94+0.015 \cdot 10}}{1 + e^{-0.94+0.015 \cdot 10}} = 0.31$$

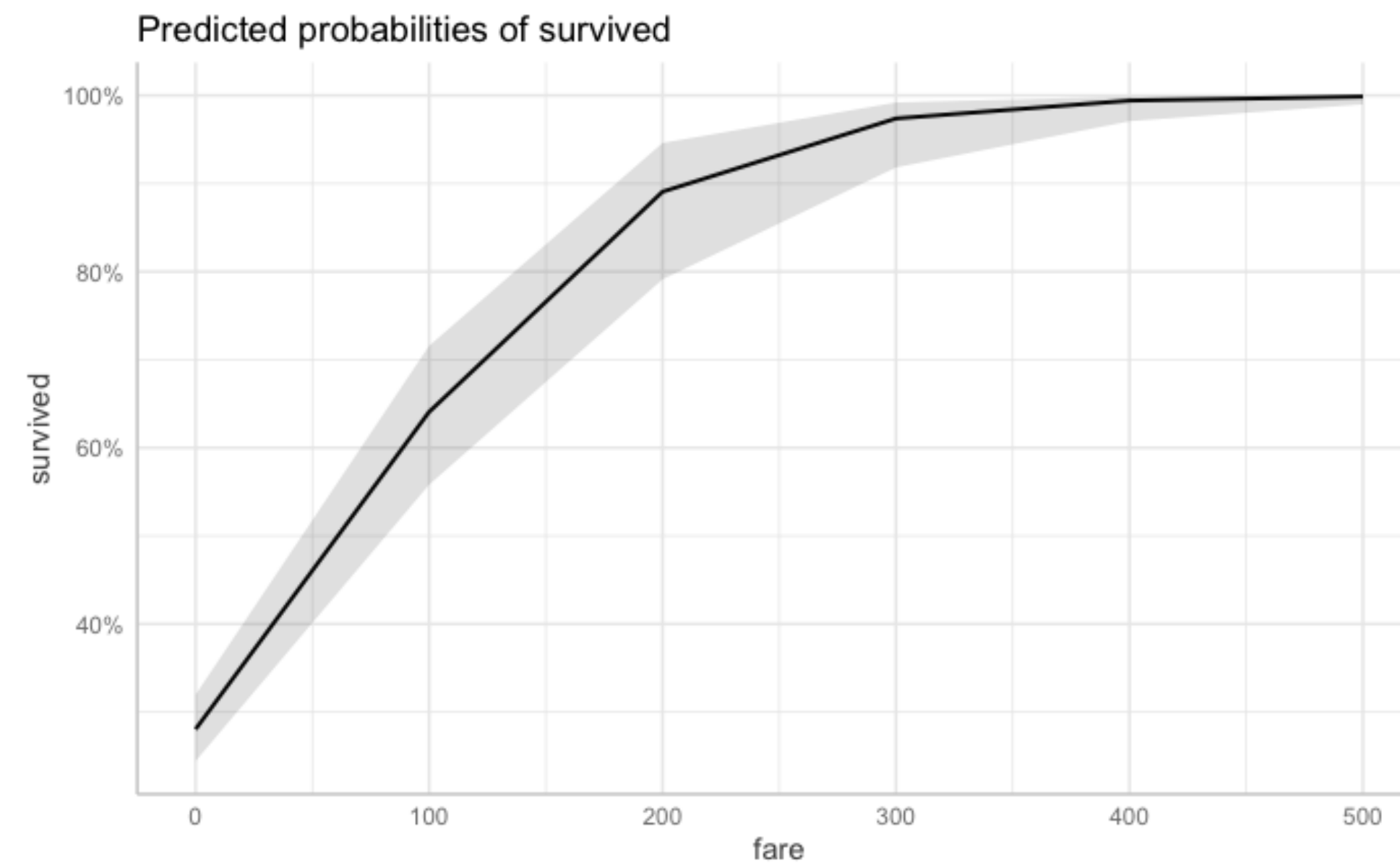
Do we have to do this by hand?



```
1 ggpredict(model = fit.glm,  
2           terms = "fare [0, 100, 200, 300, 400, 500]")
```

```
# Predicted probabilities of survived  
# x = fare
```

x	Predicted	95% CI
0	0.28	[0.24, 0.32]
100	0.64	[0.56, 0.72]
200	0.89	[0.79, 0.95]
300	0.97	[0.92, 0.99]
400	0.99	[0.97, 1.00]
500	1.00	[0.99, 1.00]



Models with several predictors

$$\ln\left(\frac{p(\text{survived})_i}{1 - p(\text{survived})_i}\right) = b_0 + b_1 \cdot \text{sex}_i + b_2 \cdot \text{fare}_i$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.647100	0.148502	4.358	1.32e-05	***
sexmale	-2.422760	0.170515	-14.208	< 2e-16	***
fare	0.011214	0.002295	4.886	1.03e-06	***

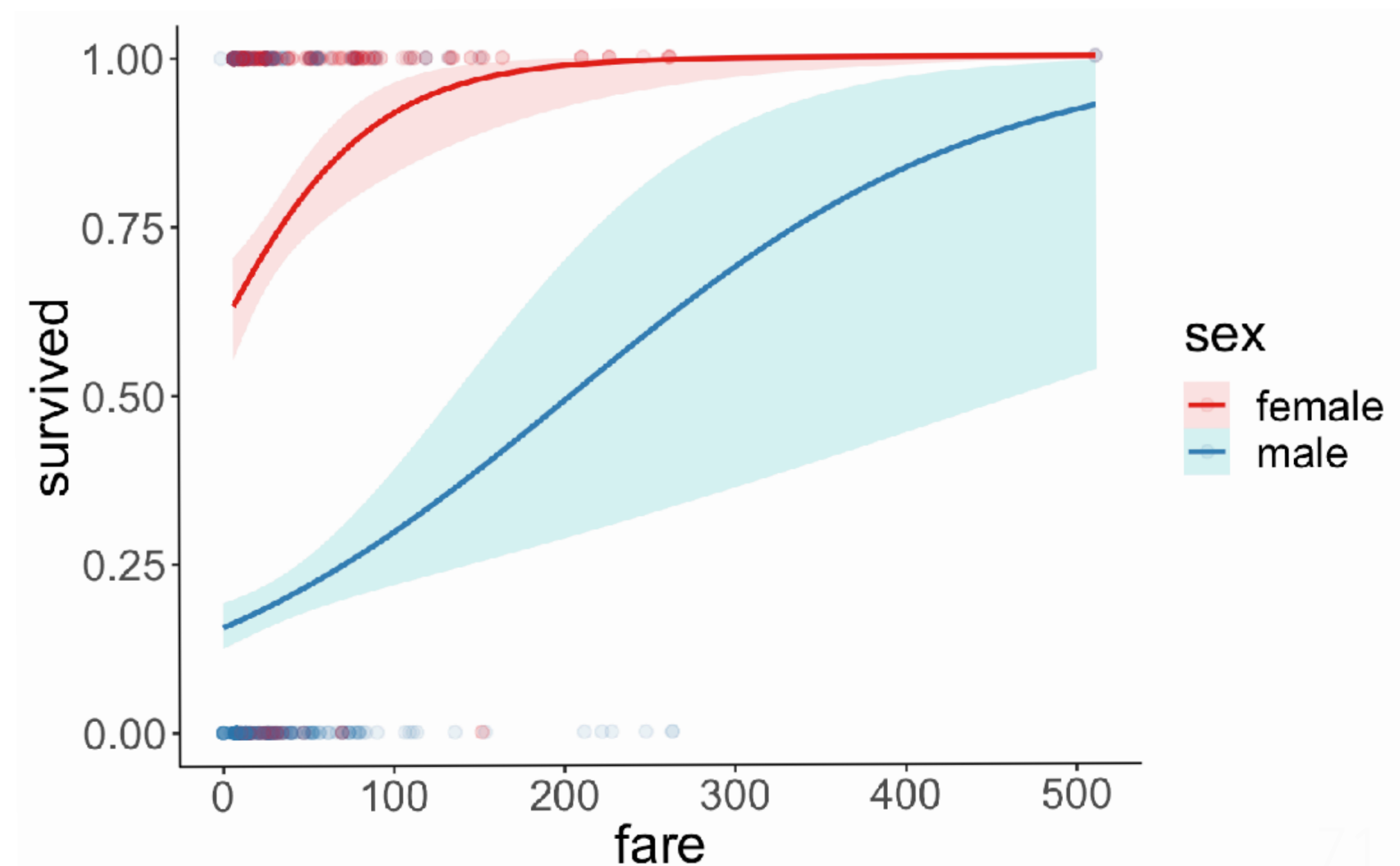
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

controlling for "fare" there is still a significant difference between female and male passengers

```
1 ggpredict(fit.glm,  
2           terms = c("sex"))
```

```
# Predicted values of survived  
# x = sex  
  
x      | Predicted | SE | 95% CI  
-----|-----|----|-----  
female | 0.73      | 0.13 | [0.68, 0.78]  
male   | 0.20      | 0.11 | [0.16, 0.23]  
  
Adjusted for:  
* fare = 32.20
```

```
1 df.titanic %>%  
2   mutate(sex = as.factor(sex)) %>%  
3   ggplot(data = ., mapping = aes(x = fare,  
4                                   y = survived,  
5                                   color = sex)) +  
6   geom_point(alpha = 0.1, size = 2) +  
7   geom_smooth(method = "glm",  
8               method.args = list(family = "binomial"),  
9               alpha = 0.2,  
10              aes(fill = sex)) +  
11   scale_color_brewer(palette = "Set1")
```



Fitting and reporting models

Simulating a logistic regression

```
1 # make example reproducible
2 set.seed(1)
3
4 # set parameters
5 sample_size = 1000
6 b0 = 0
7 b1 = 1
8
9 # generate data
10 df.data = tibble(
11   x = rnorm(n = sample_size),
12   y = b0 + b1 * x,
13   p = inv.logit(y))
14   mutate(response = rbinom(n(), size = 1, p = p))
15
16 # fit model
17 fit = glm(formula = response ~ 1 + x,
18           family = "binomial",
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20
21 # model summary
22 fit %>% summary()
```

set some parameters

linear model (y is in log odds)

transform into probability

randomly draw response

fit a logistic regression

summarize the result

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22 fit %>% summary()
```

```
Call:
glm(formula = response ~ 1 + x, family = "binomial", data = df.data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.1137  -1.0118  -0.4591   1.0287   2.2591

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.06214    0.06918  -0.898    0.369
x            0.92905    0.07937  11.705 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1385.4  on 999  degrees of freedom
Residual deviance: 1209.6  on 998  degrees of freedom
AIC: 1213.6

Number of Fisher Scoring iterations: 3
```

Assessing the model fit

actual value

predicted value

$$\text{log-likelihood} = \sum_{i=1}^n [Y_i \cdot \ln(P(Y_i)) + (1 - Y_i) \cdot \ln(1 - P(Y_i))]$$

- calculate the probability of the observed response
- take the log of these probabilities
- sum them up to get the log-likelihood of the data (given the model)

Clue guide to probability

- **joint probability:**
- if A and B are independent then
- **Definition:** $p(A, B) = p(A) \cdot p(B)$
- $p(\text{Prof Plum, candle stick}) = p(\text{Prof Plum}) \cdot p(\text{candle stick}) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

Who?

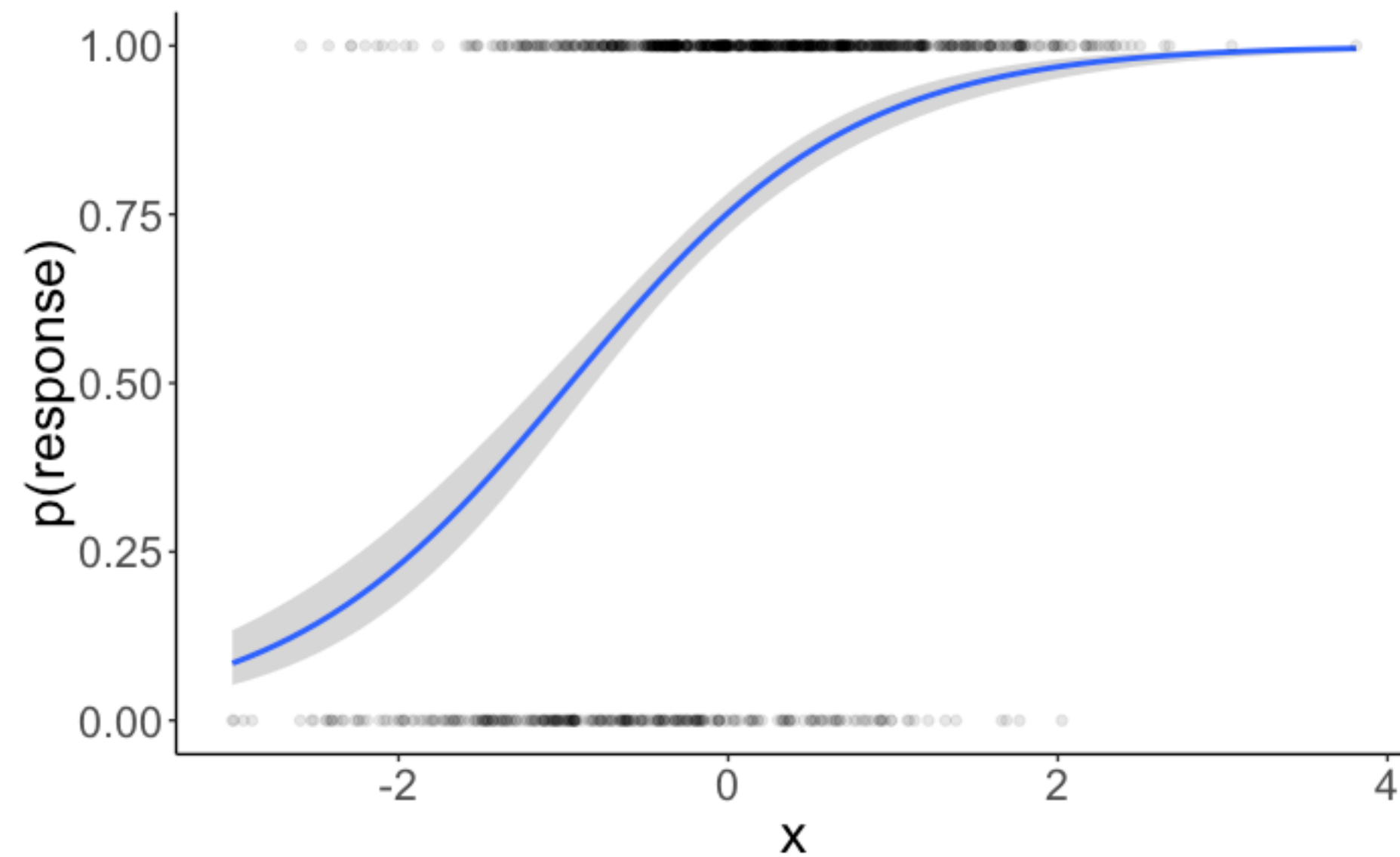
What?

41

response	p(Y = 1)	p(Y = response)	log(p(Y = response))
1	0.34	0.34	-1.07
0	0.53	0.47	-0.75
1	0.30	0.30	-1.20
1	0.81	0.81	-0.22
1	0.56	0.56	-0.58
0	0.30	0.70	-0.36
1	0.60	0.60	-0.52
1	0.65	0.65	-0.43
1	0.62	0.62	-0.48
0	0.41	0.59	-0.54

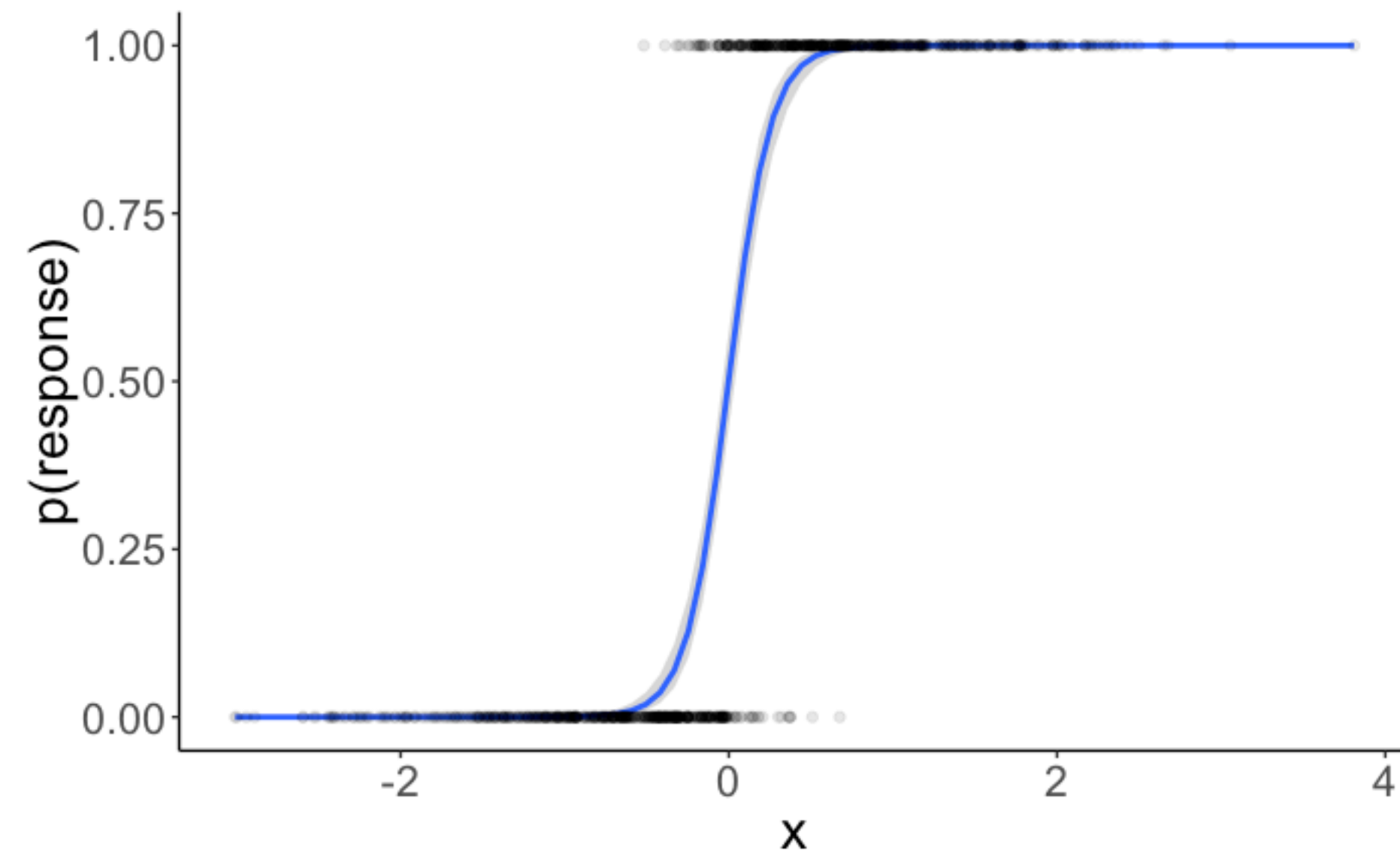
Assessing the model fit

doesn't predict the
response very well



logLik	AIC	BIC
-501.65	1007.3	1017.12

predicts the response
much better



logLik	AIC	BIC
-156.37	316.74	326.55

Testing hypotheses

aka checking
whether it's **worth it**

```
1 # fit compact model
2 fit.compact = glm(formula = survived ~ 1 + fare,
3                   family = "binomial",
4                   data = df.titanic)
5
6 # fit augmented model
7 fit.augmented = glm(formula = survived ~ 1 + sex + fare,
8                    family = "binomial",
9                    data = df.titanic)
10
11 # likelihood ratio test
12 anova(fit.compact, fit.augmented, test = "LRT")
```

**we need to specify that we
want a likelihood ratio test**

Analysis of Deviance Table

Model 1: survived ~ 1 + fare

Model 2: survived ~ 1 + sex + fare

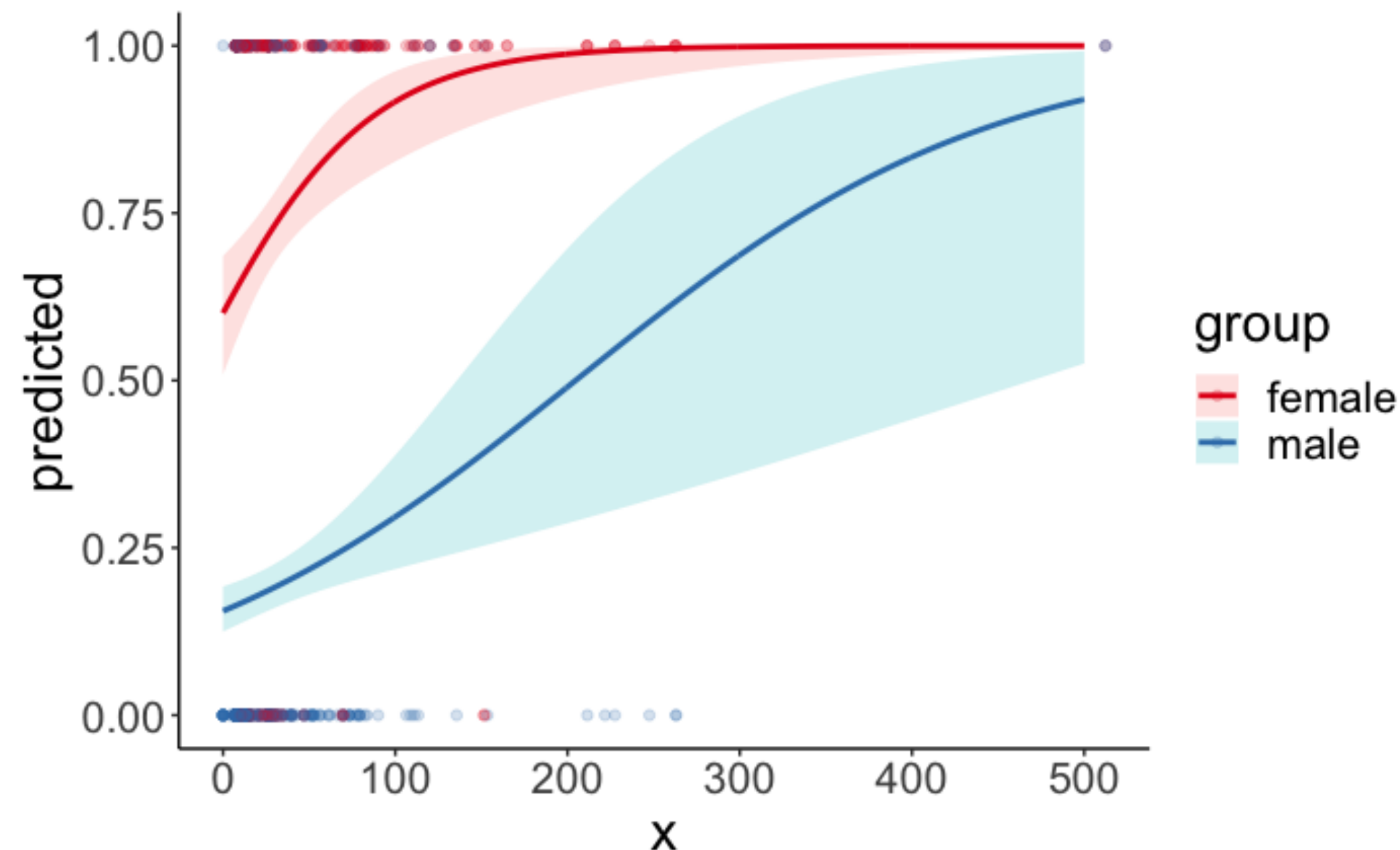
	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	889	1117.57			
2	888	884.31	1	233.26	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Reporting results



- Visualize the data
- Show a table with the regression results
- Report significance of different factors
- Interpreting parameter estimates is tricky -- probably best to report probabilities for a few example cases



```
# Predicted values of survived
# x = fare

# sex = female
```

x	Predicted	SE	95% CI
0	0.60	0.19	[0.51, 0.69]
100	0.92	0.42	[0.83, 0.96]
200	0.99	0.95	[0.93, 1.00]
300	1.00	1.48	[0.97, 1.00]
400	1.00	2.02	[0.99, 1.00]
500	1.00	2.55	[1.00, 1.00]

```
# sex = male
```

x	Predicted	SE	95% CI
0	0.16	0.13	[0.12, 0.19]
100	0.30	0.21	[0.22, 0.39]
200	0.49	0.44	[0.29, 0.70]
300	0.69	0.69	[0.36, 0.90]
400	0.83	0.94	[0.44, 0.97]
500	0.92	1.19	[0.53, 0.99]

Assumptions

- linearity (between predictors and log odds)
- independence
- no multi-collinearity
- model fails to converge when there is **complete separation**:
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Different kinds of generalized models

Different linking functions

```
binomial(link = "logit")  
  
gaussian(link = "identity")  
  
Gamma(link = "inverse")  
  
inverse.gaussian(link = "1/mu^2")  
  
poisson(link = "log")  
  
quasi(link = "identity", variance = "constant")  
  
quasibinomial(link = "logit")  
  
quasipoisson(link = "log")
```

**apply different transformations to the
dependent variable**

Fitting and reporting models

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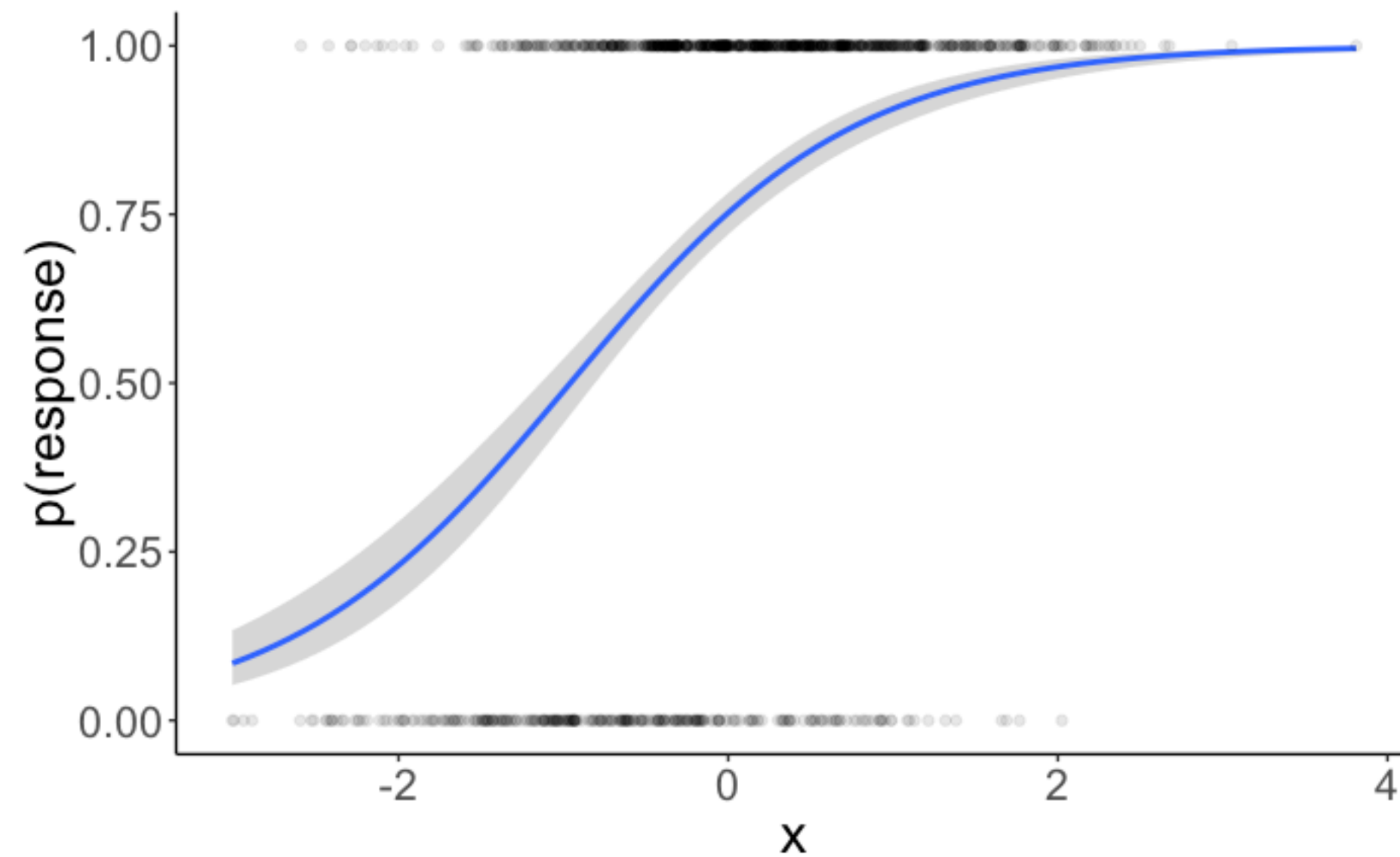
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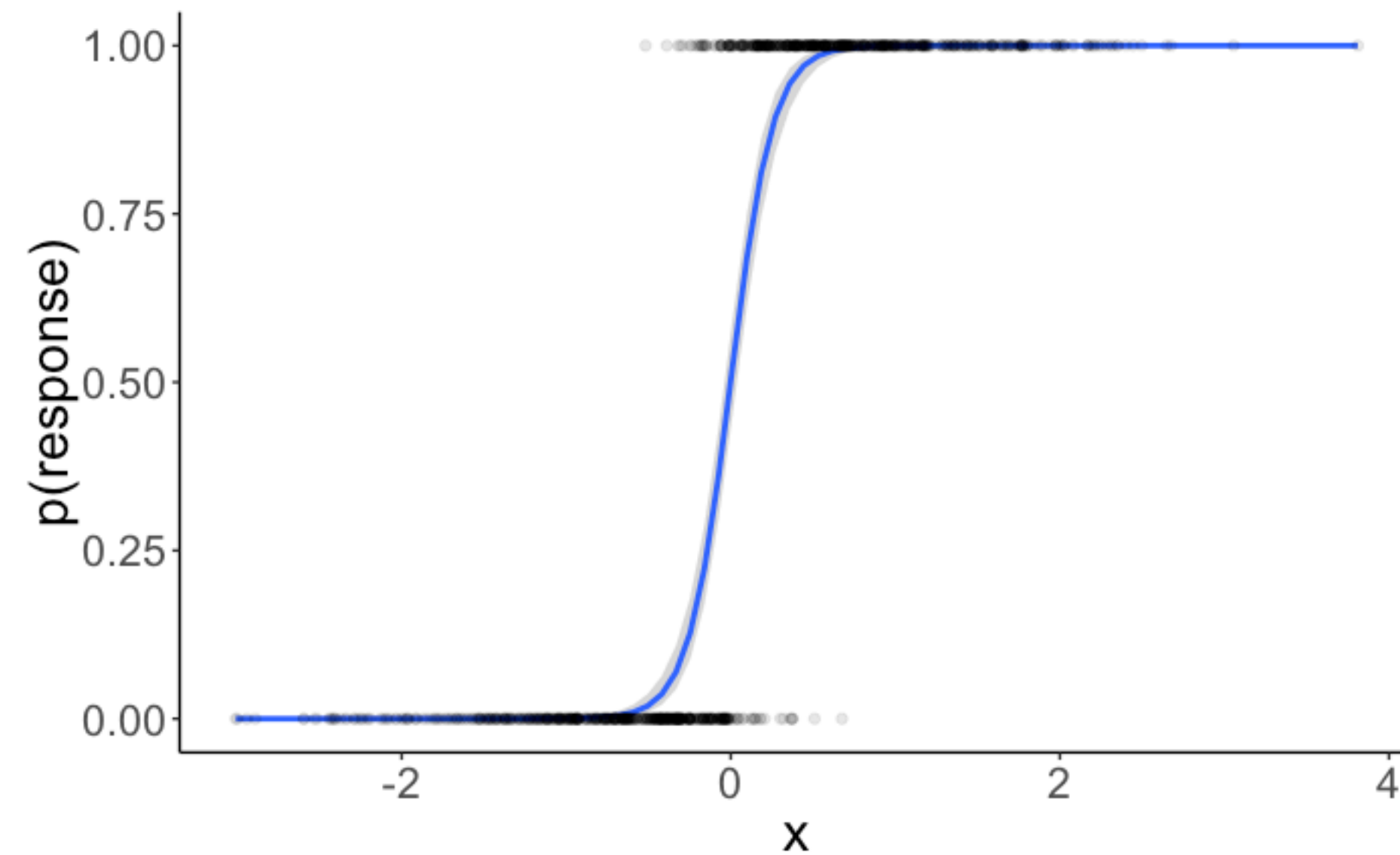
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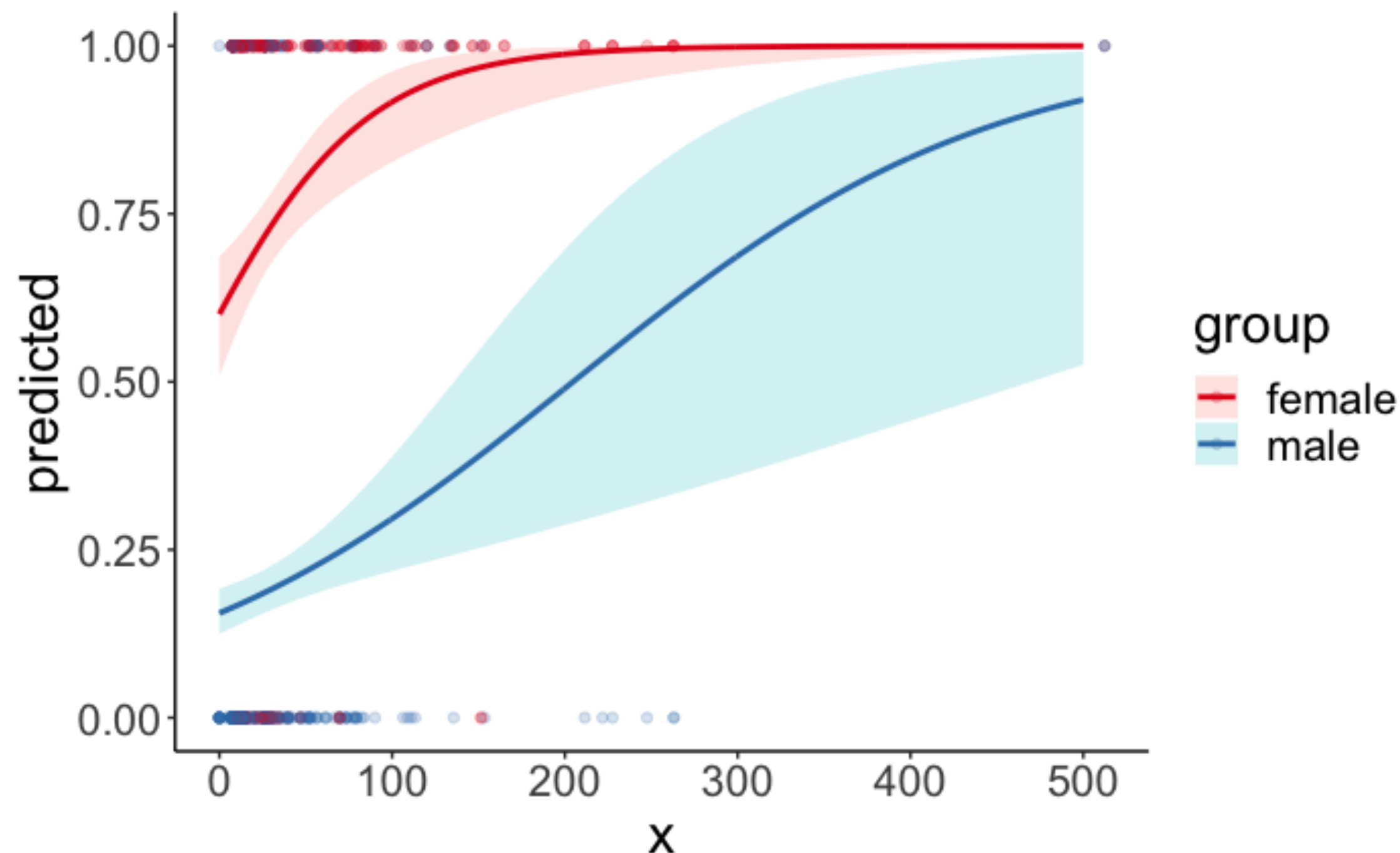
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```

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Plan for today

- Quick Recap
- Interpreting parameters
- Unbalanced designs
- Linear contrasts
- Generalized linear model
 - Logistic regression
 - interpreting the model output
 - fitting and reporting models

Feedback

How was the pace of today's class?

much too slow a little too slow just right a little too fast much too fast

How happy were you with today's class overall?



What did you like about today's class? What could be improved next time?

Thank you!